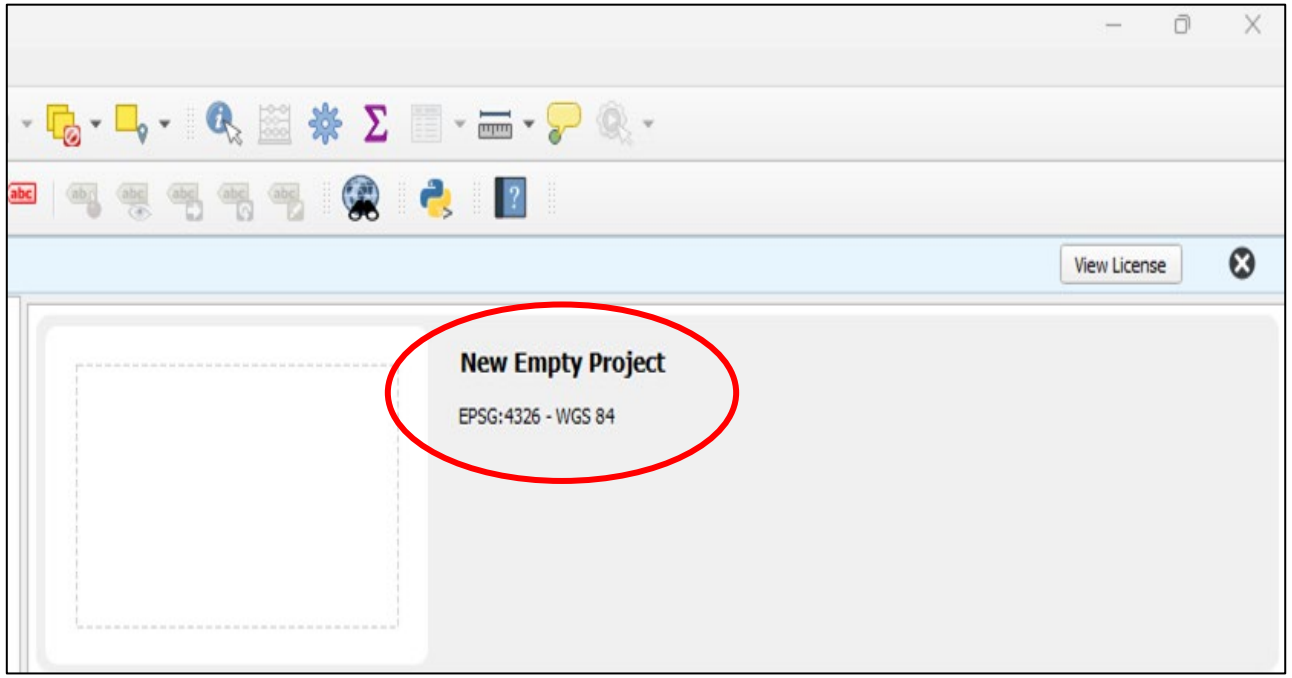
Start *QGIS*

Download workshop data

- Extract /unzip the .zip file
- Save it where you can find it…

open QGIS (your version may look different)

- Double click on *New Empty Project*



Note: New *QGIS* projects open with Geographic Coordinate System (GCS) **EPSG:4326**

Understand *QGIS* interface

*your interface may look different

Menu Bar

Toolbar

Layers Panel

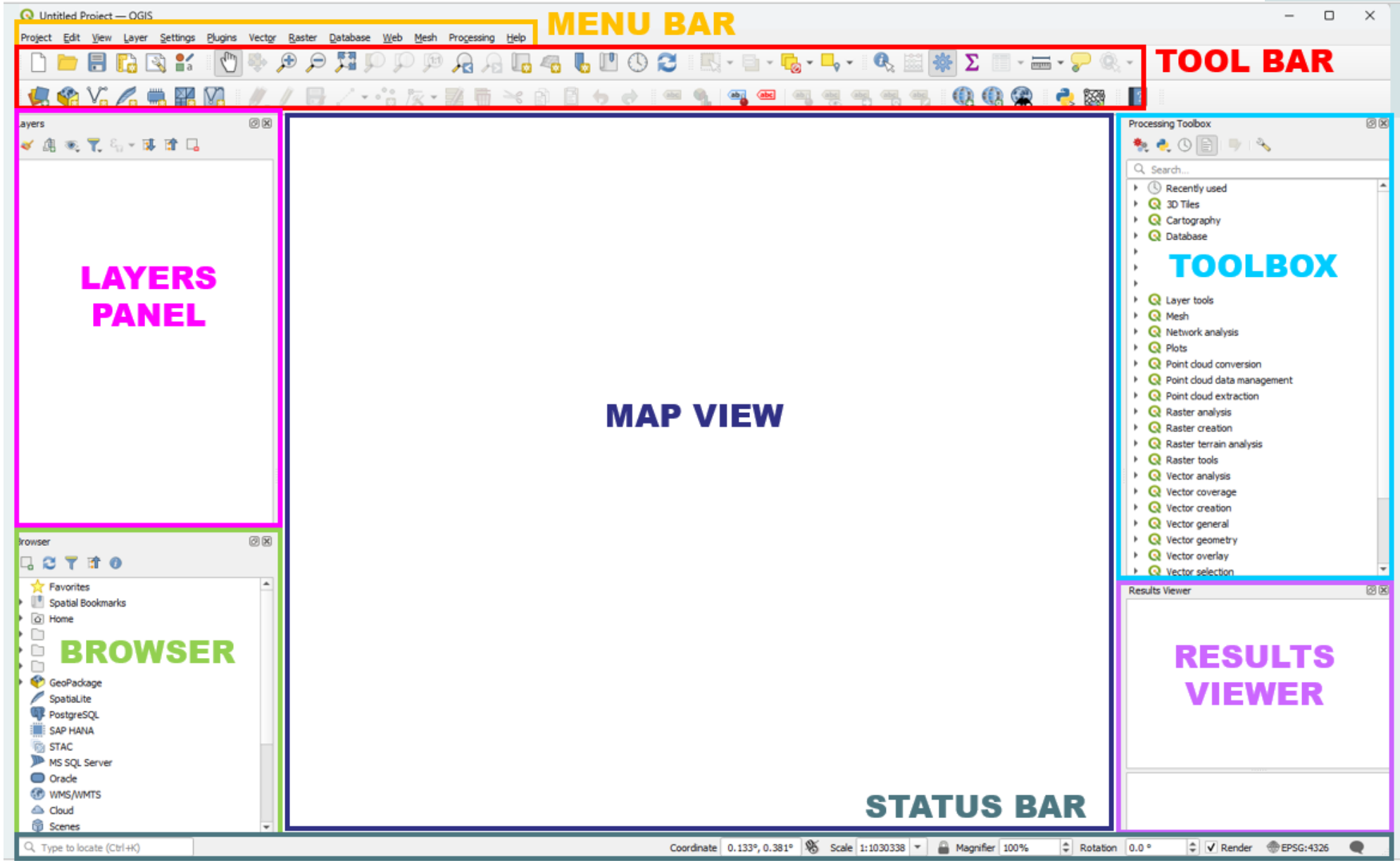
Map view

Browser

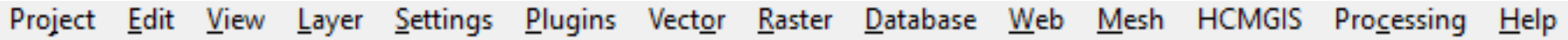
Toolbox

Status Bar

Results Viewer



Understand *QGIS* interface

Menu Bar: 

Horizontal bar at the top providing access to various functions and tools

(Project management, Edit, Plugins, Vector & Raster tools, etc.)

Toolbar:

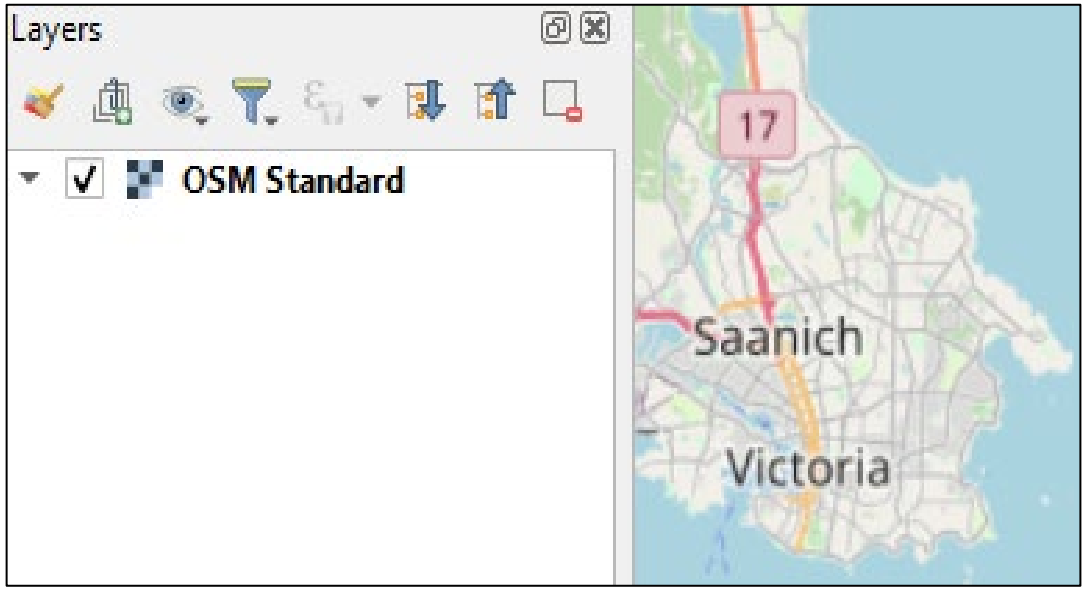
Contains icons for frequently used tools, such as Add Data, Pan, Zoom, Identify, etc.

Quick access to essential operations.

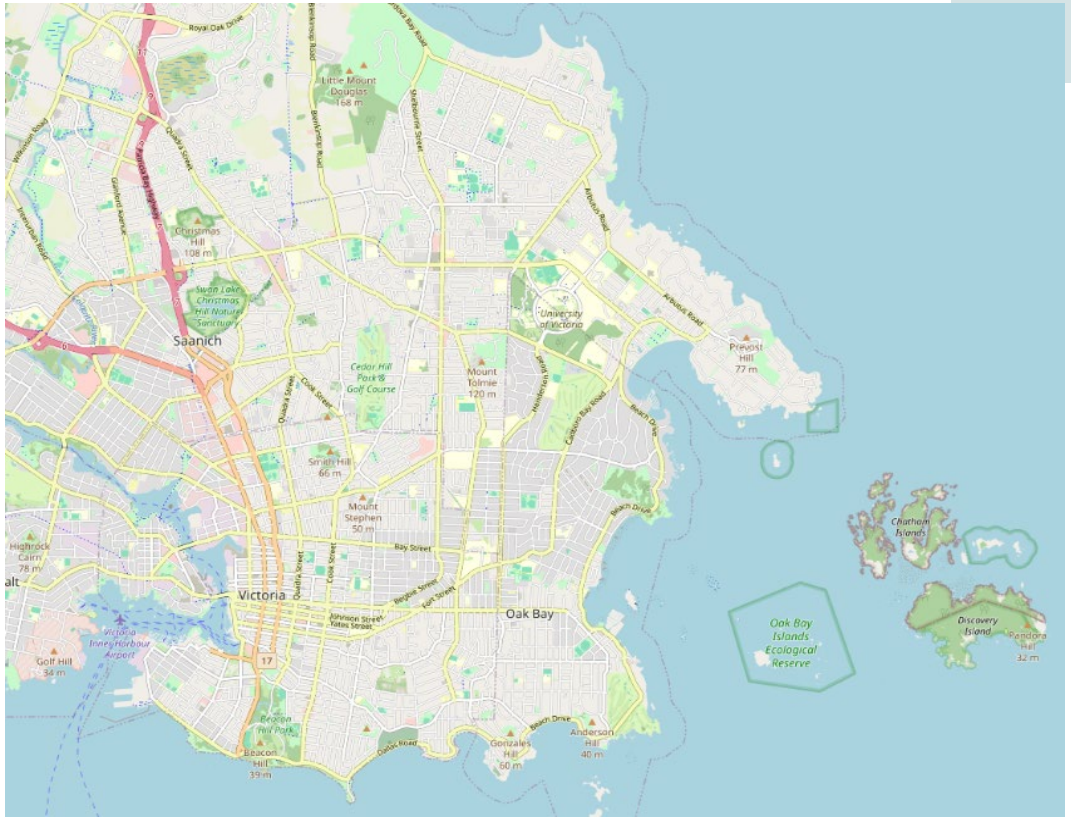
Layers Panel:

Displays all active **Layers** in the project.

Allows users to organize, manage visibility and access properties of layers



Understand *QGIS* interface

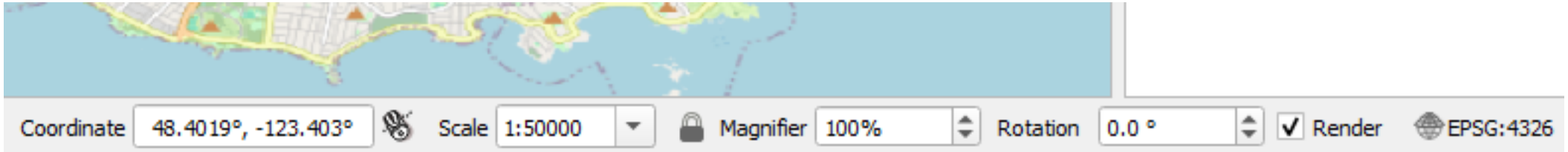


Map View:

The central area where spatial data is displayed. Interact with the map, visualize layers and analyze spatial relationships.

Status Bar:

Located at the bottom, it provides information about current project: coordinate display, scale and CRS settings.



Toolbar essentials

New project, open, and save

Pan

zoom

“**zoom full**”

Add data

Activity #1

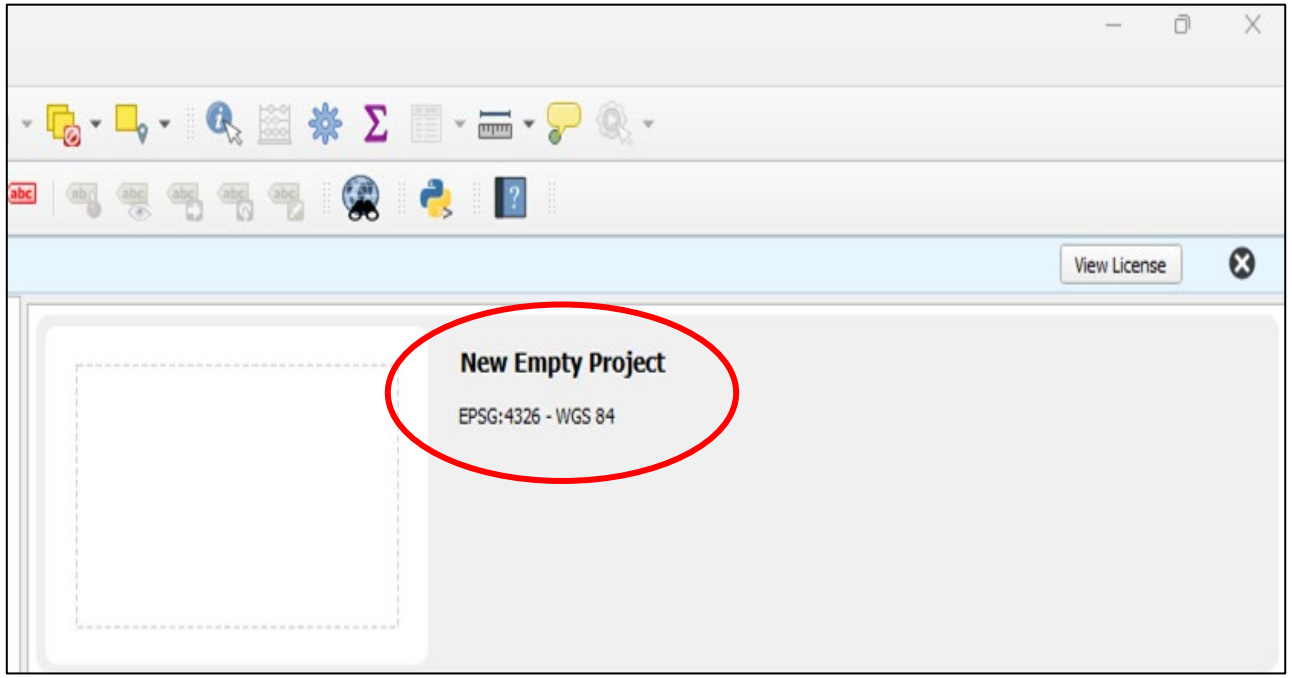
Start *QGIS*

Download workshop data

- Extract /unzip the .zip file
- Save it where you can find it…

Open QGIS (your version may look different)

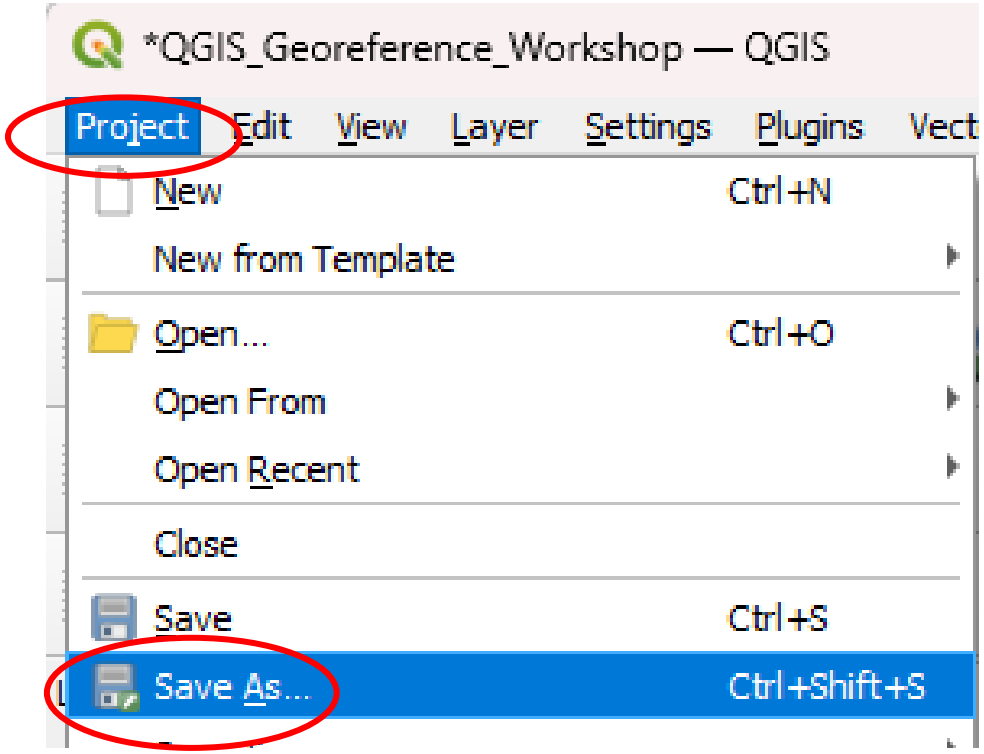
- Double click on *New Empty Project*

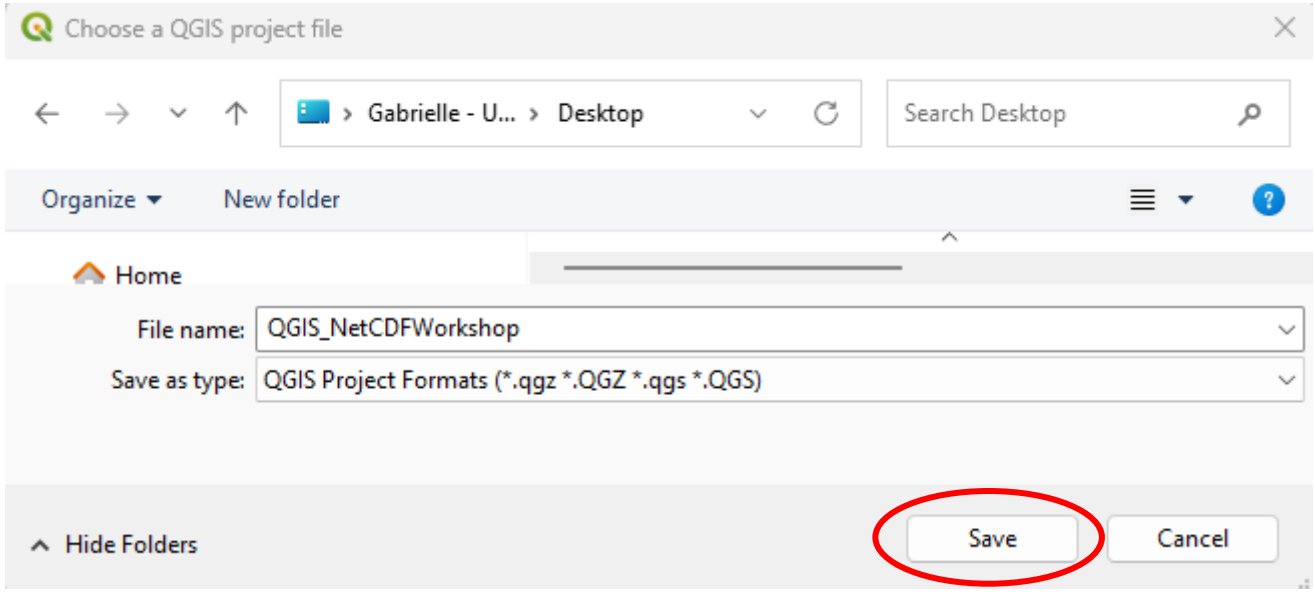


Note: New *QGIS* projects open with Geographic Coordinate System (GCS) **EPSG:4326**

Save new project

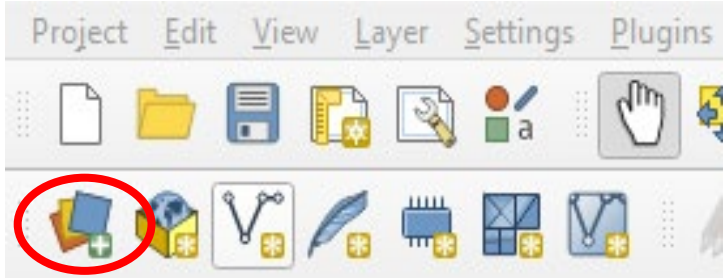
- In *QGIS* Menu Bar, select *Project* then *Save As*
- Name your project "**QGIS_NetCDFWorkshop**"
- Save your project as **.qgz** to where you can find it

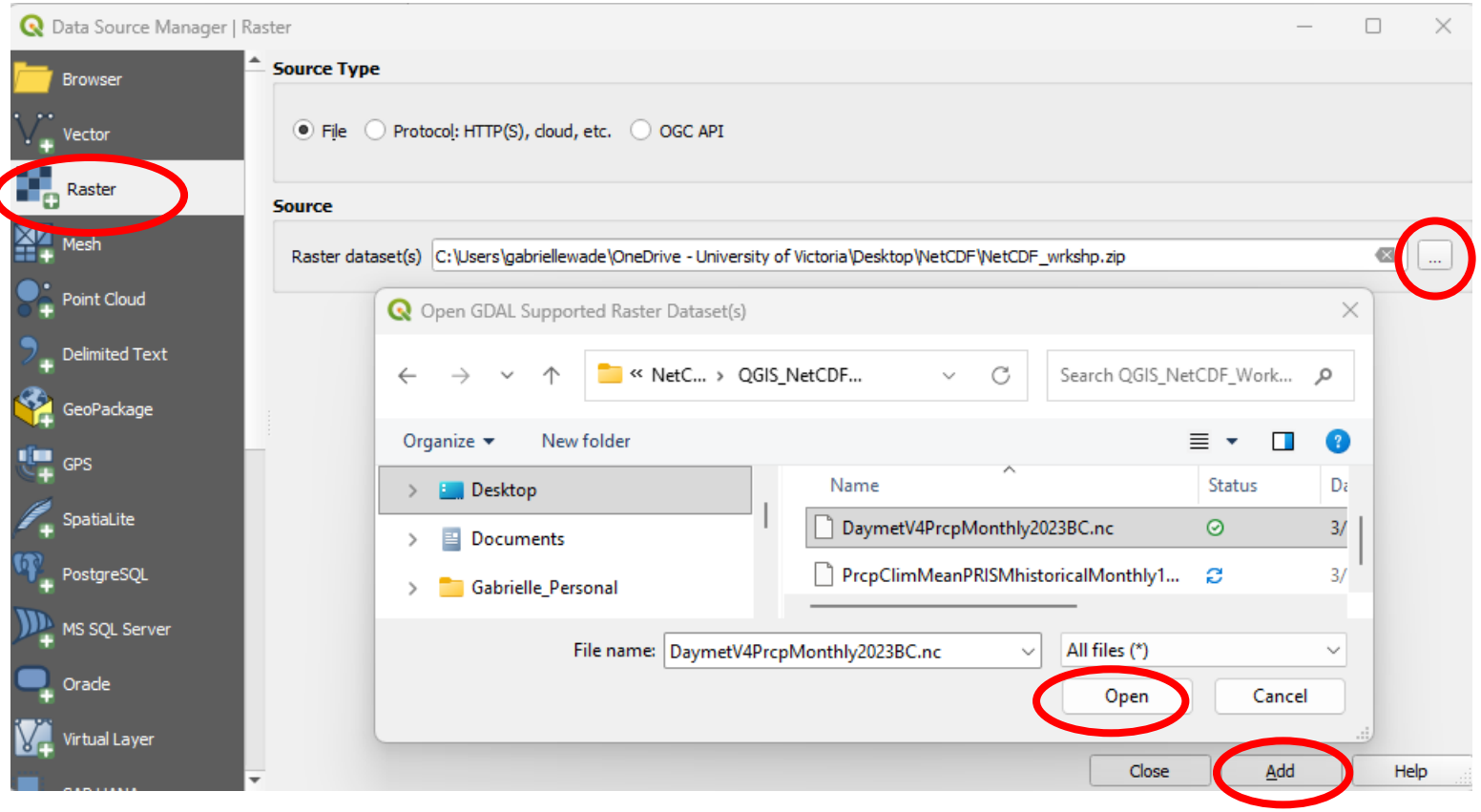




Note: .qgz is the project file format for *QGIS*

add NetCDF Layer

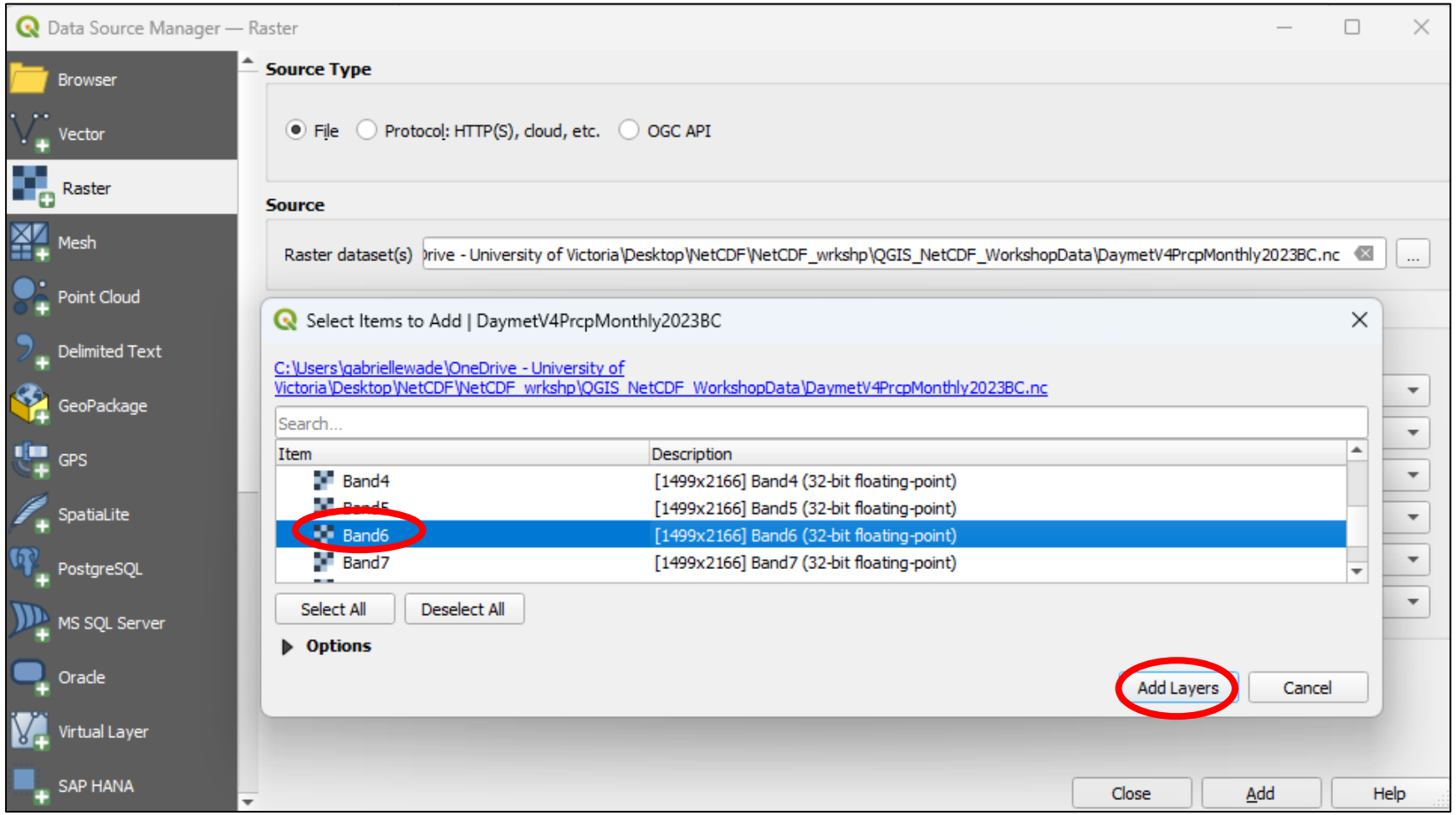




- Select *Open Data Source Manager*
- Select the *Raster* tab
- Under the "Source" heading click the
- Navigate to workshop data
- Select **DaymetV4PrcpMonthly2023BC.nc,** Open
- **Add**

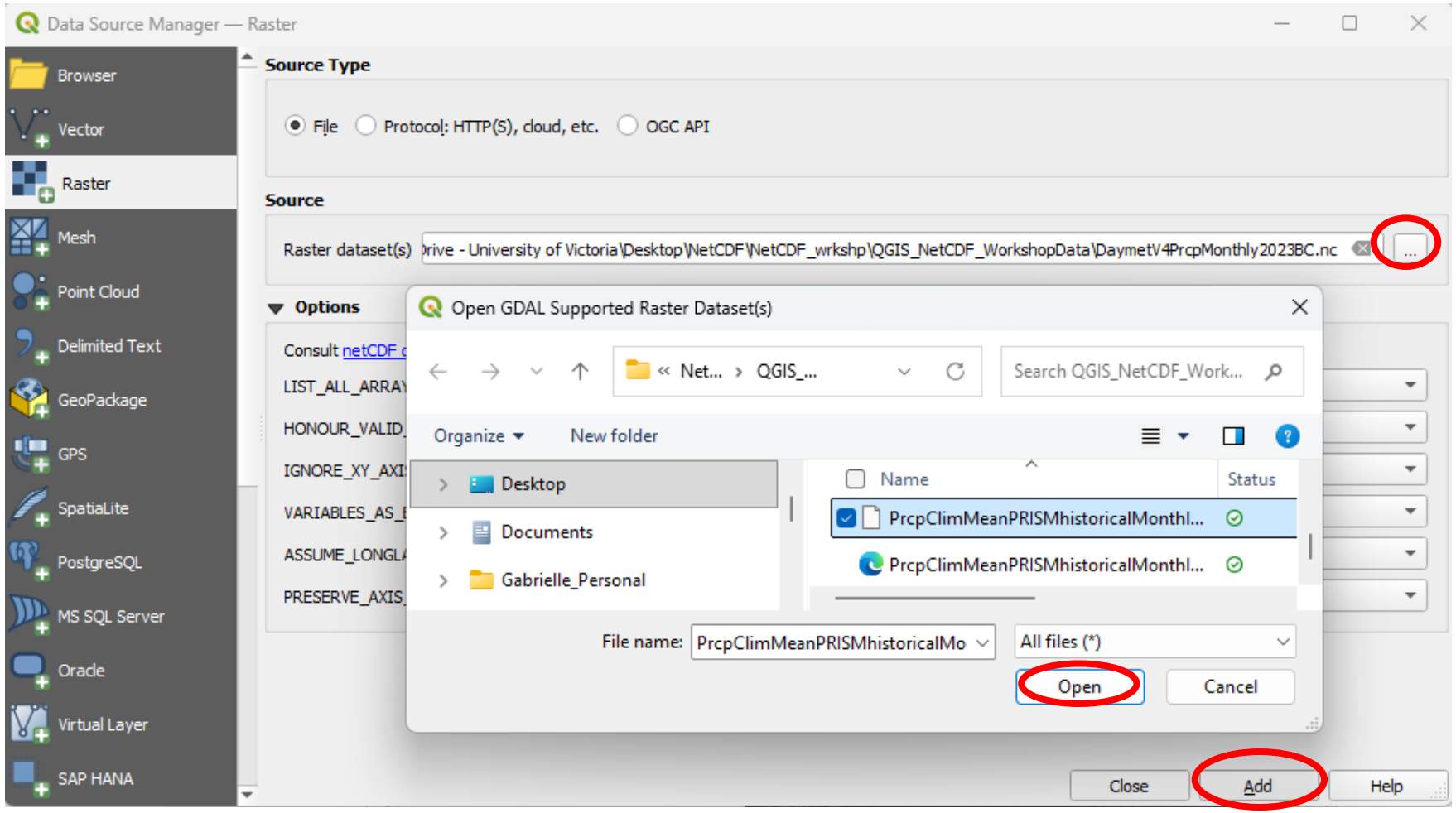
add NetCDF Layer

- Pop-up will ask to “Select Items to Add”
- Bands represent months of the year (Band 1 = Jan, Band 2 = Feb, etc.)
    - NetCDF can hold many temporal layers or other variables (temp, etc.)
- add **Band 6** to get monthly precipitation for June 2023 (the month before fire started)
- *Add Layers*



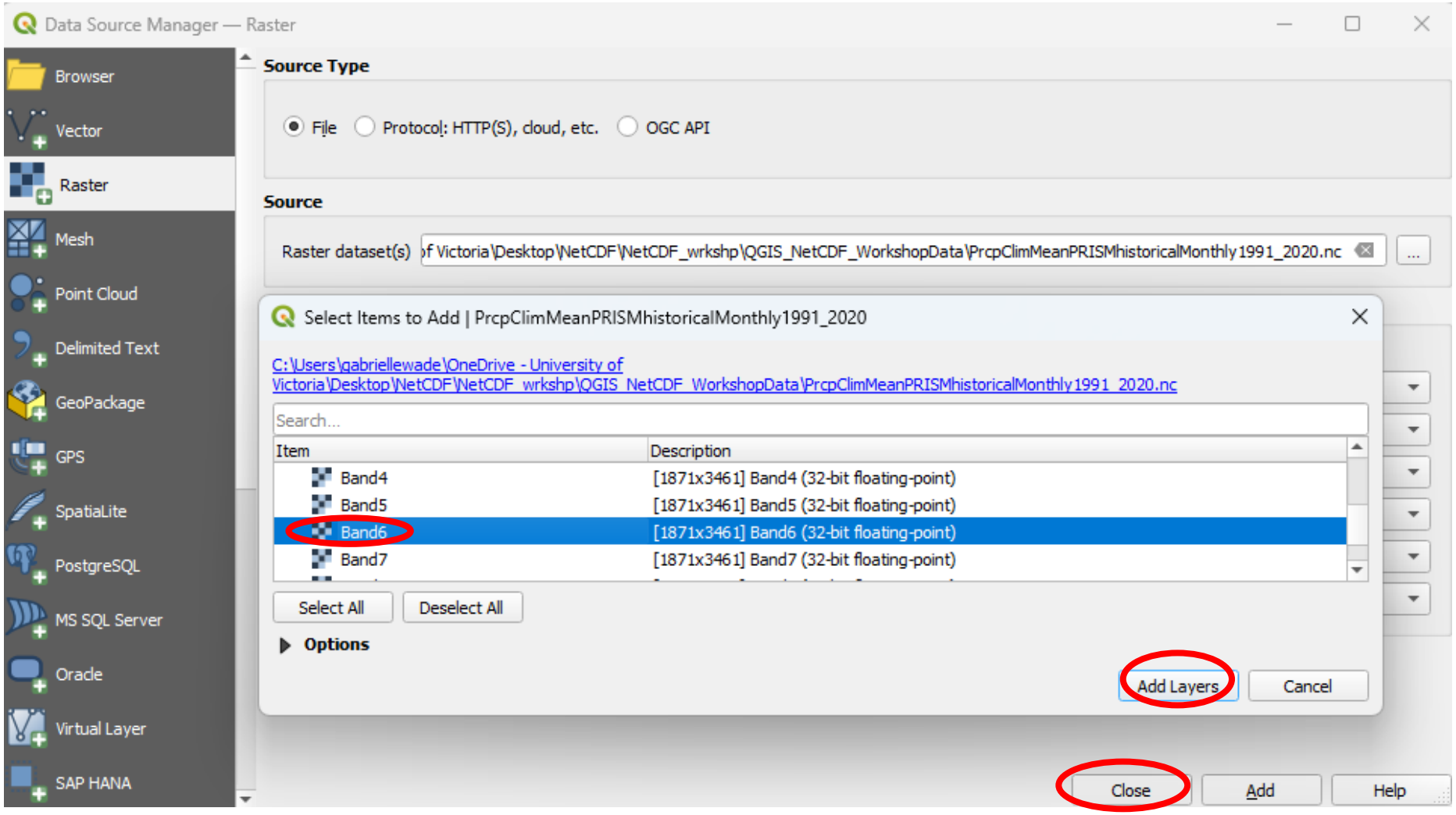
add (another) NetCDF Layer

- Click the 3 dots under *Source* heading again
- Navigate to workshop data
- Select **PrcpClimMeanPRISMhistoricalMonthly1991_2020.nc;** Open
- **Add**



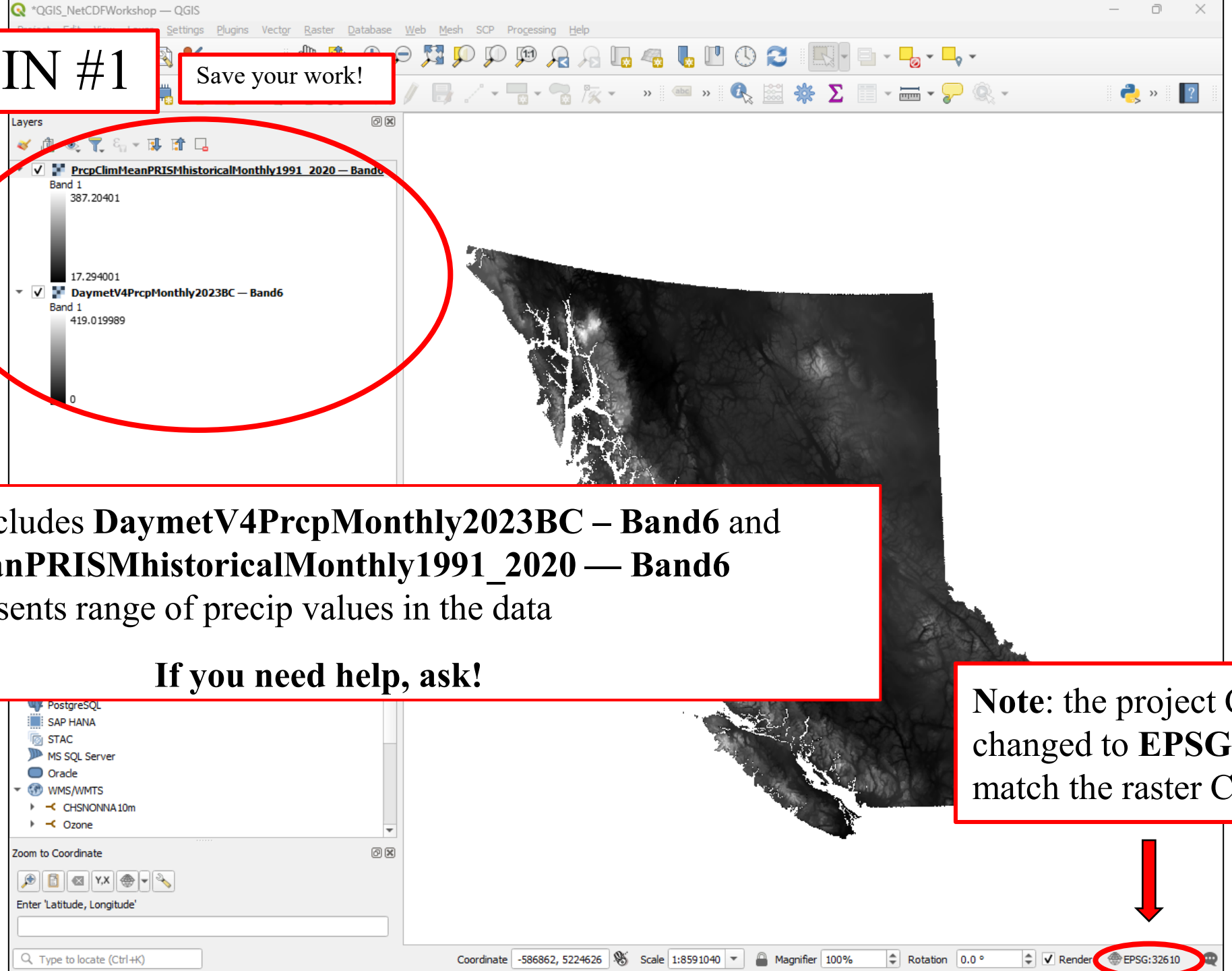
add (another) NetCDF Layer

- Pop-up will ask to “Select Items to Add”
- add **Band 6** to get monthly normal precipitation for 1991-2020 June
- *Add Layers* and *Close*



CHECK IN #1

Save your work!



- *Layers* now includes **DaymetV4PrcpMonthly2023BC – Band6** and **PrcpClimMeanPRISMhistoricalMonthly1991_2020 — Band6**
- Gradient represents range of precip values in the data

If you need help, ask!

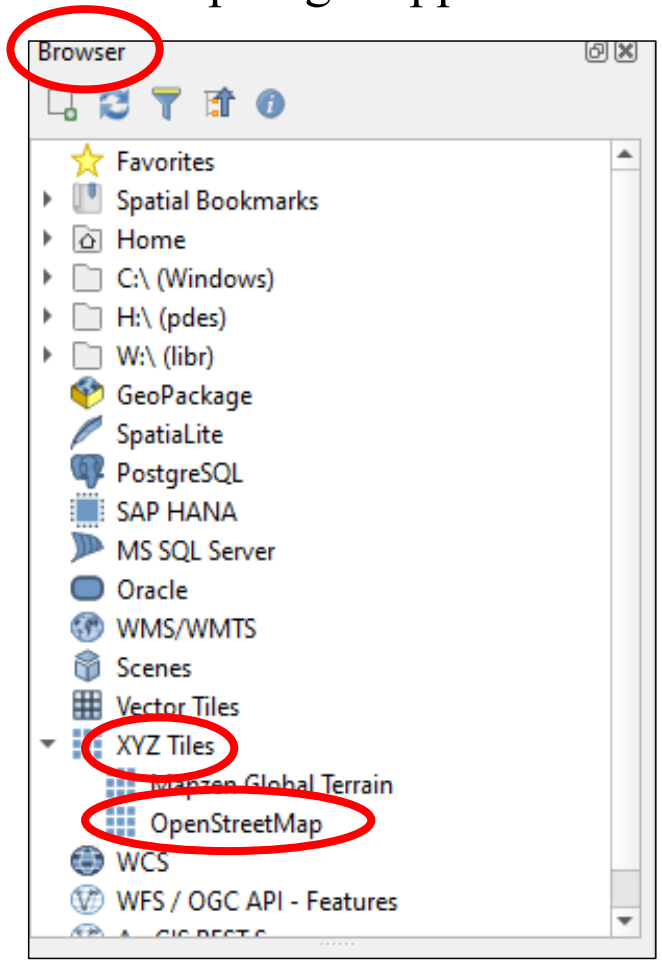
Note: the project CRS has changed to **EPSG:32610** to match the raster CRS

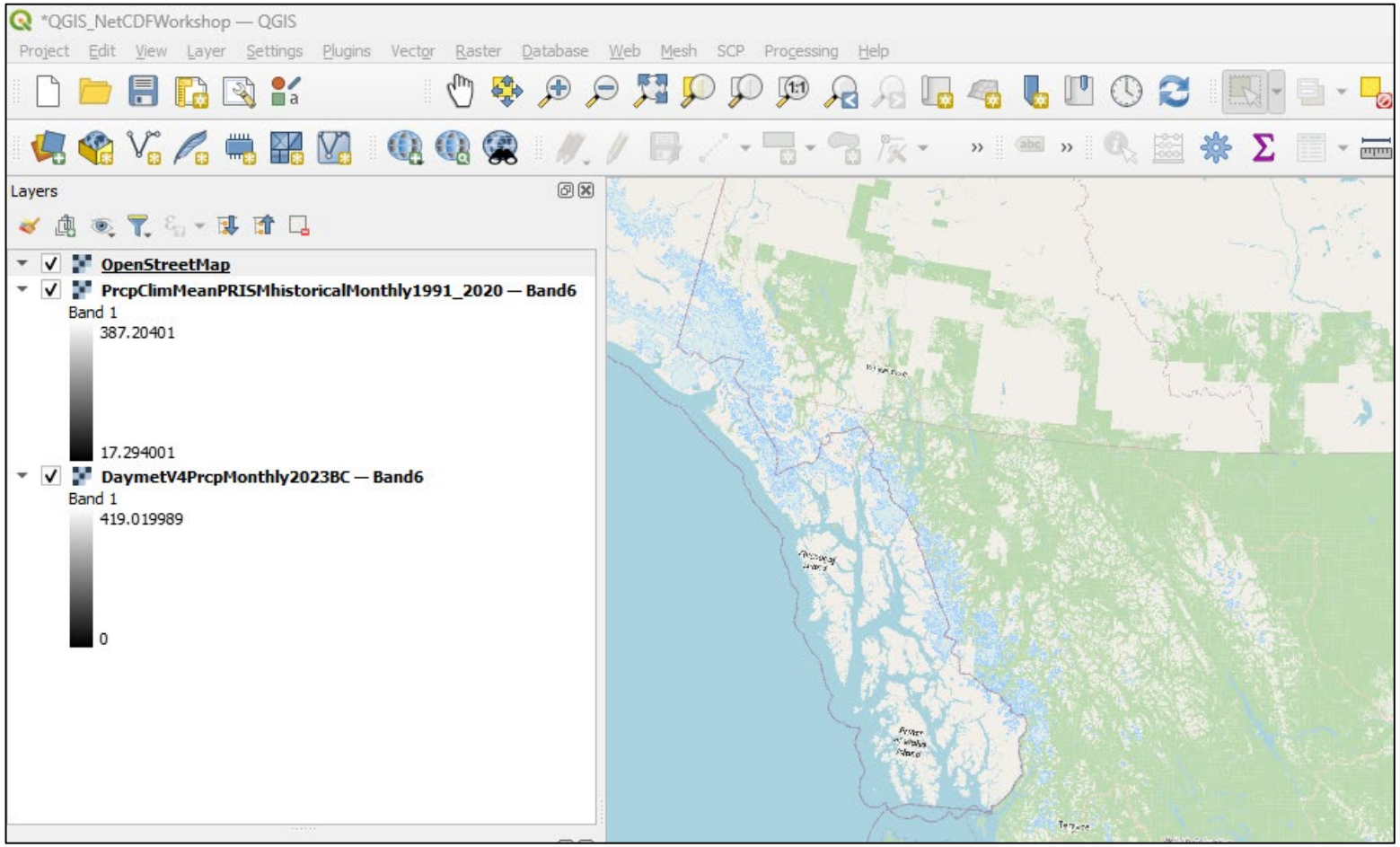
Activity #2

add a Basemap

Add a **basemap** for location context

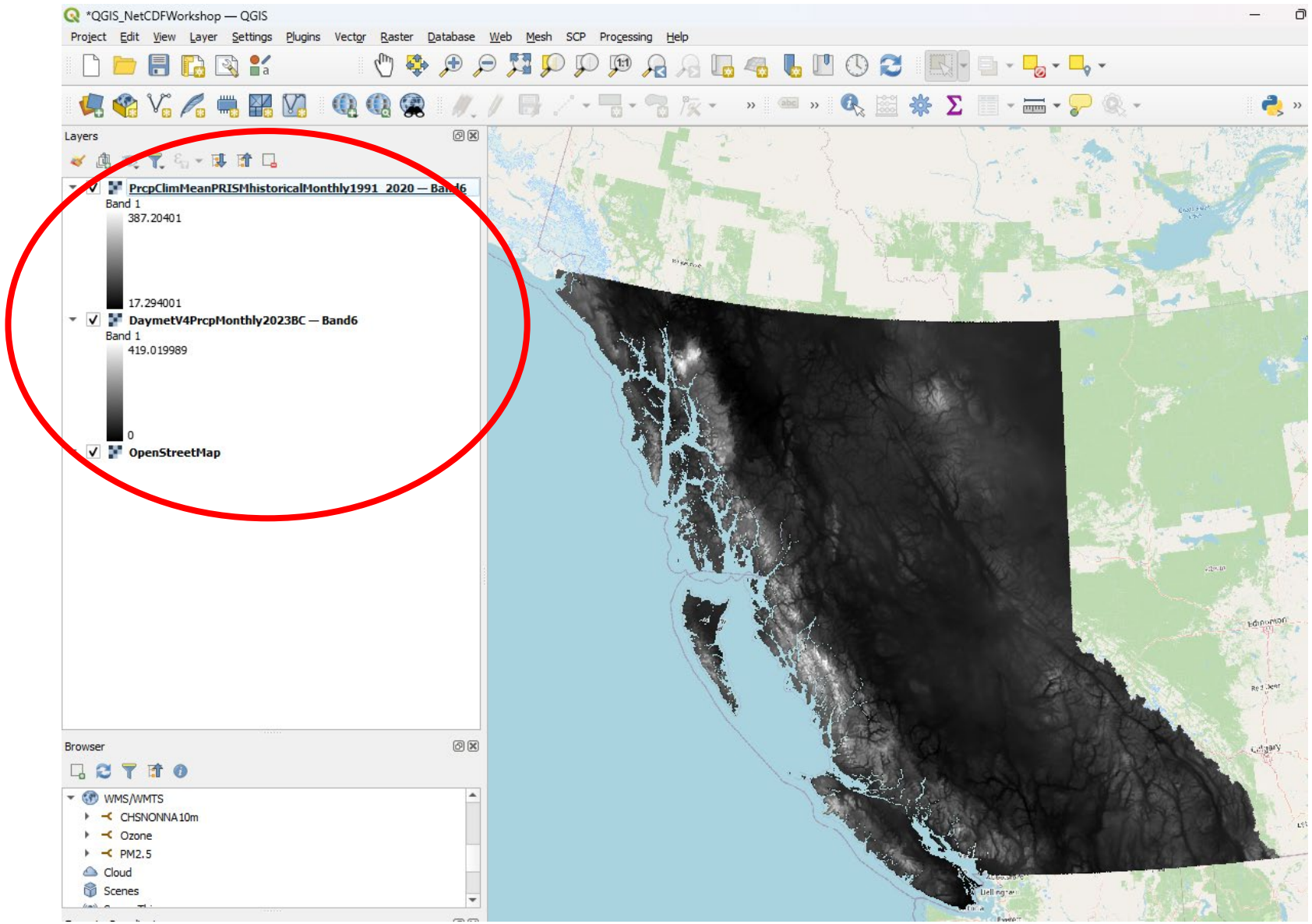
- in the ‘Browser’, expand *XYZ Tiles*
- double-click *OpenStreetMap* to add to map (if a warning appears, press OK)
- Basemap might appear above NetCDF layers in *Layers Panel* and *Map View*





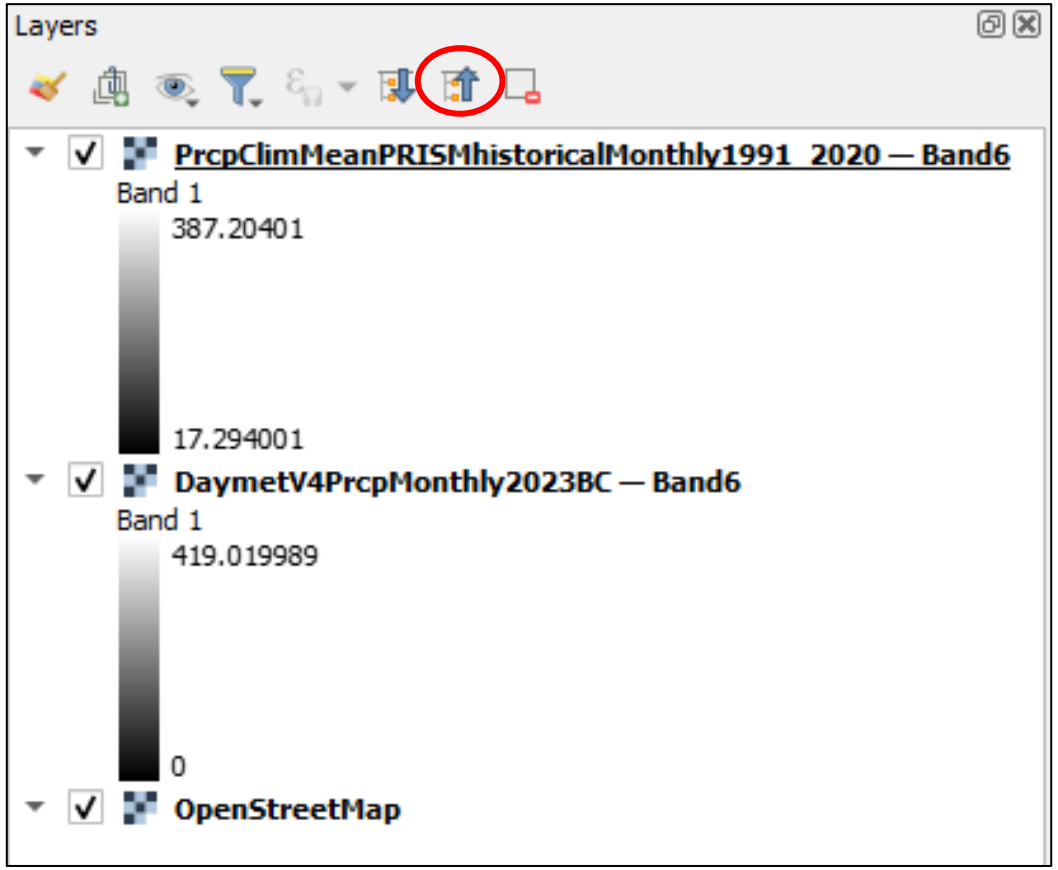
arrange layers

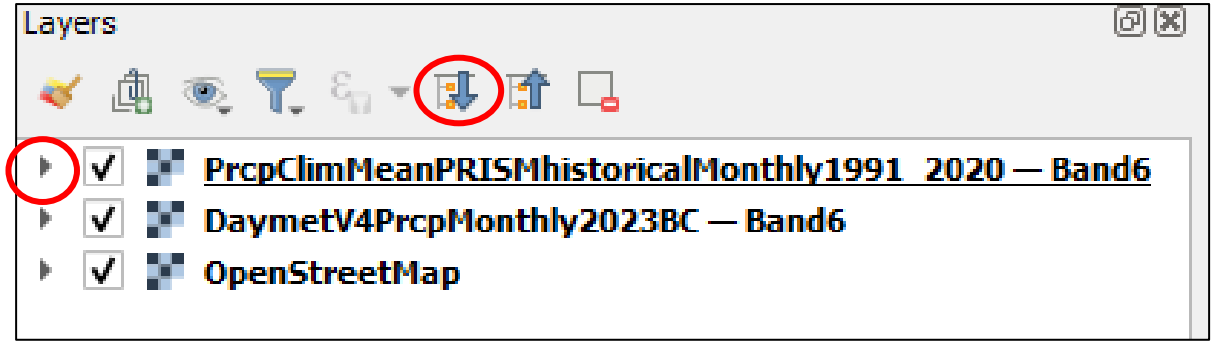
- Click on **OpenStreetMap** in *Layers* panel and drag below NetCDF layers



organize *Layers* panel

- collapse NetCDF layers in *Layers* panel to look ‘better’
- select *Collapse All*

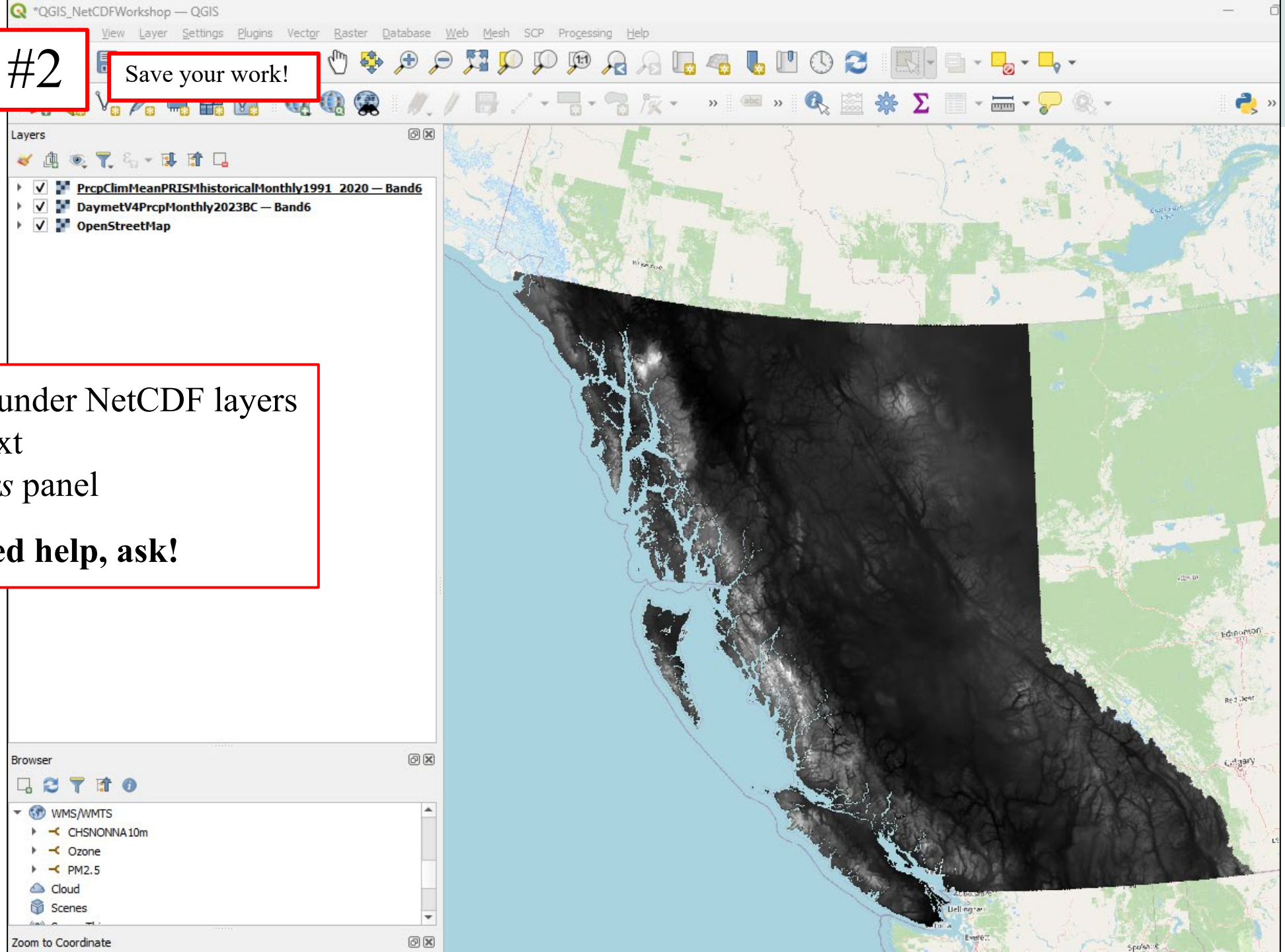




can (re)expand raster layers using *Expand all* or drop-down arrows

CHECK IN #2

Save your work!



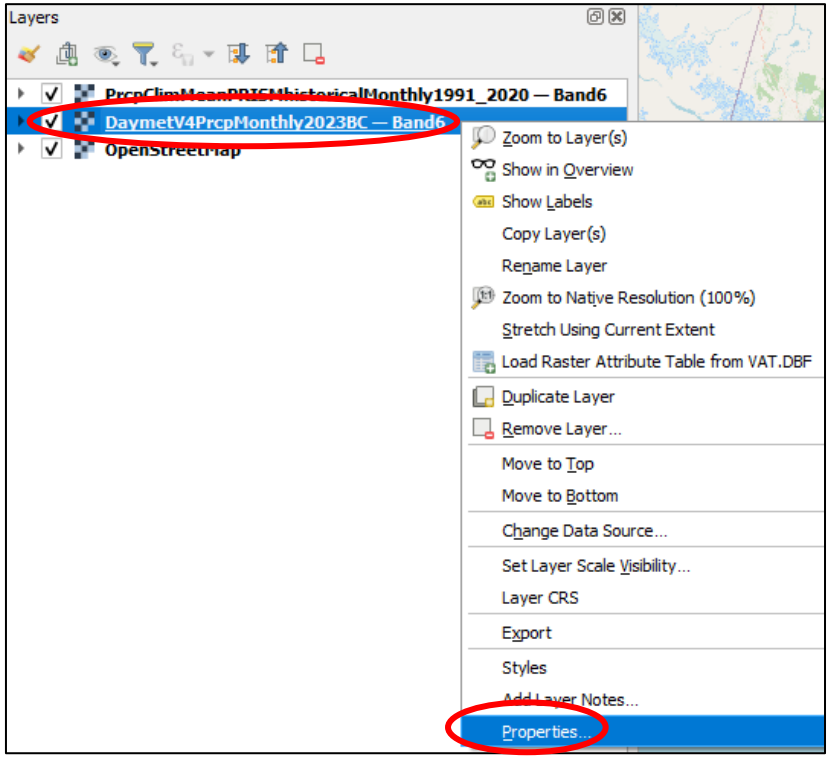
- added a basemap under NetCDF layers for location context
- re-arranged *Layers* panel

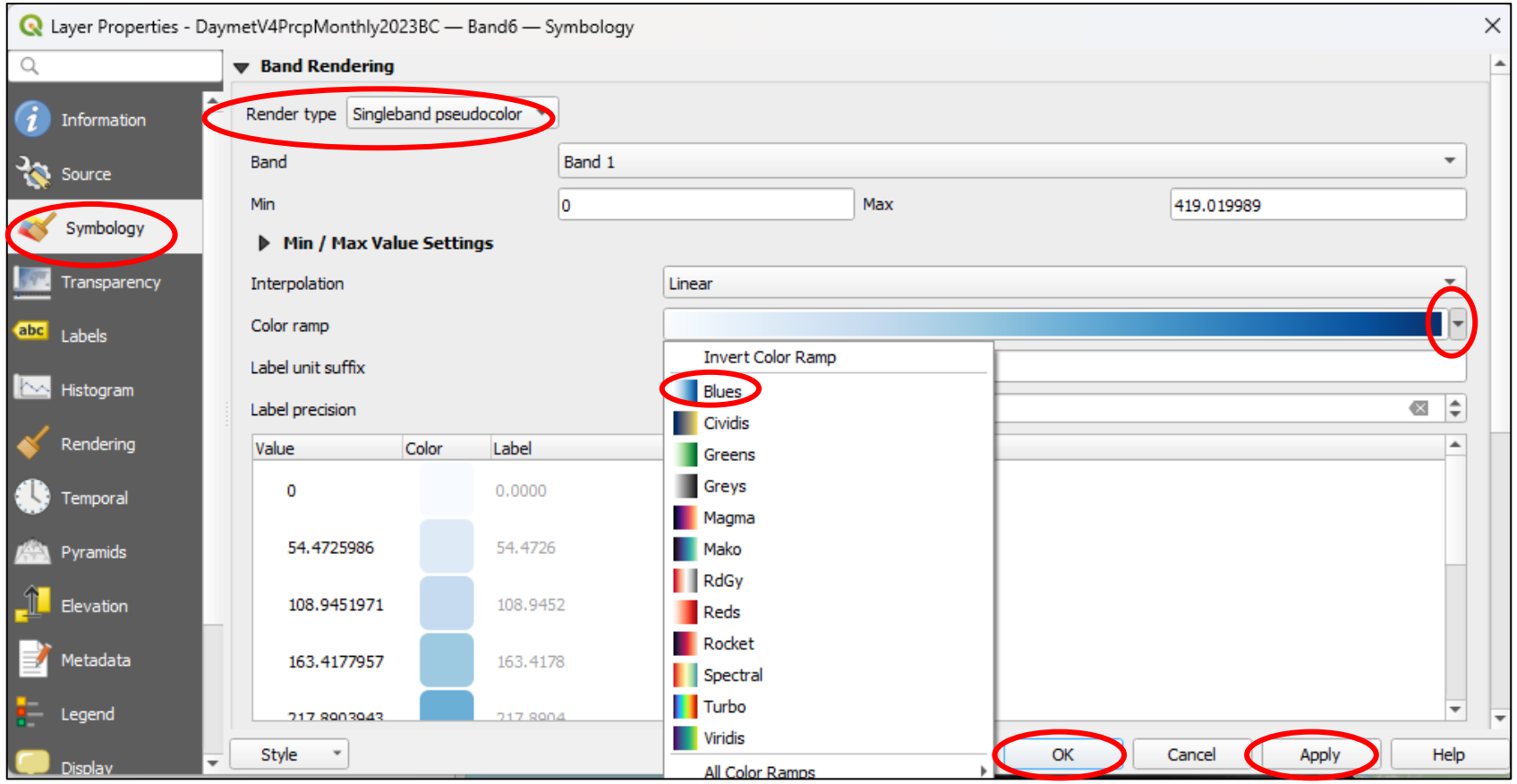
If you need help, ask!

Activity #3

change NetCDF Symbology

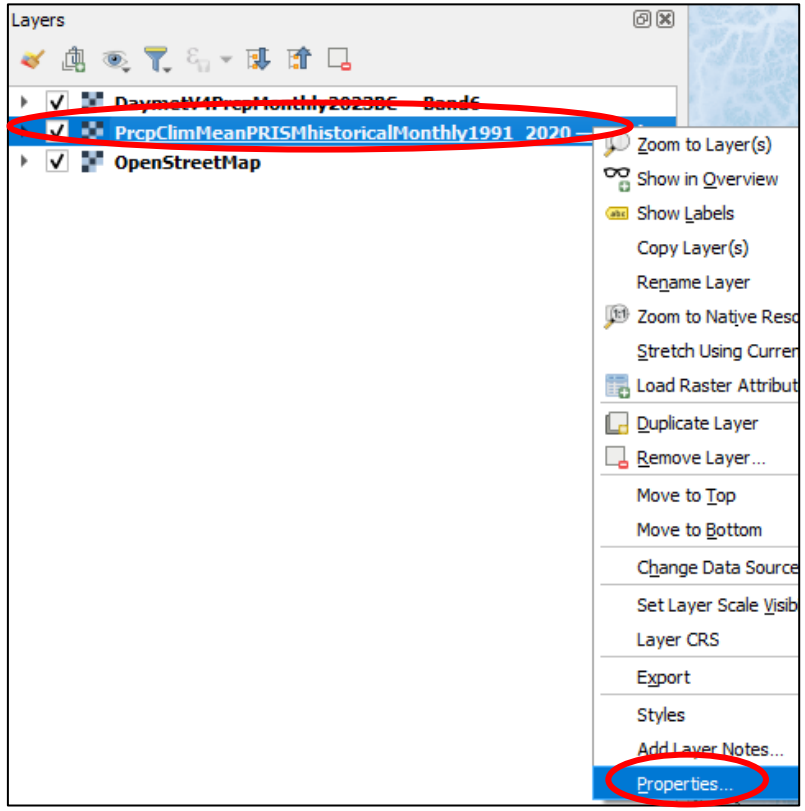
- right-click on **DaymetV4PrcpMonthly2023BC — Band6**, select *Properties*
- under *Symbology,* change *Render type* to “Singleband pseudocolor” in drop-down
- use drop-down arrow beside *Color ramp* to choose “Blues”
- **Apply** and **OK**

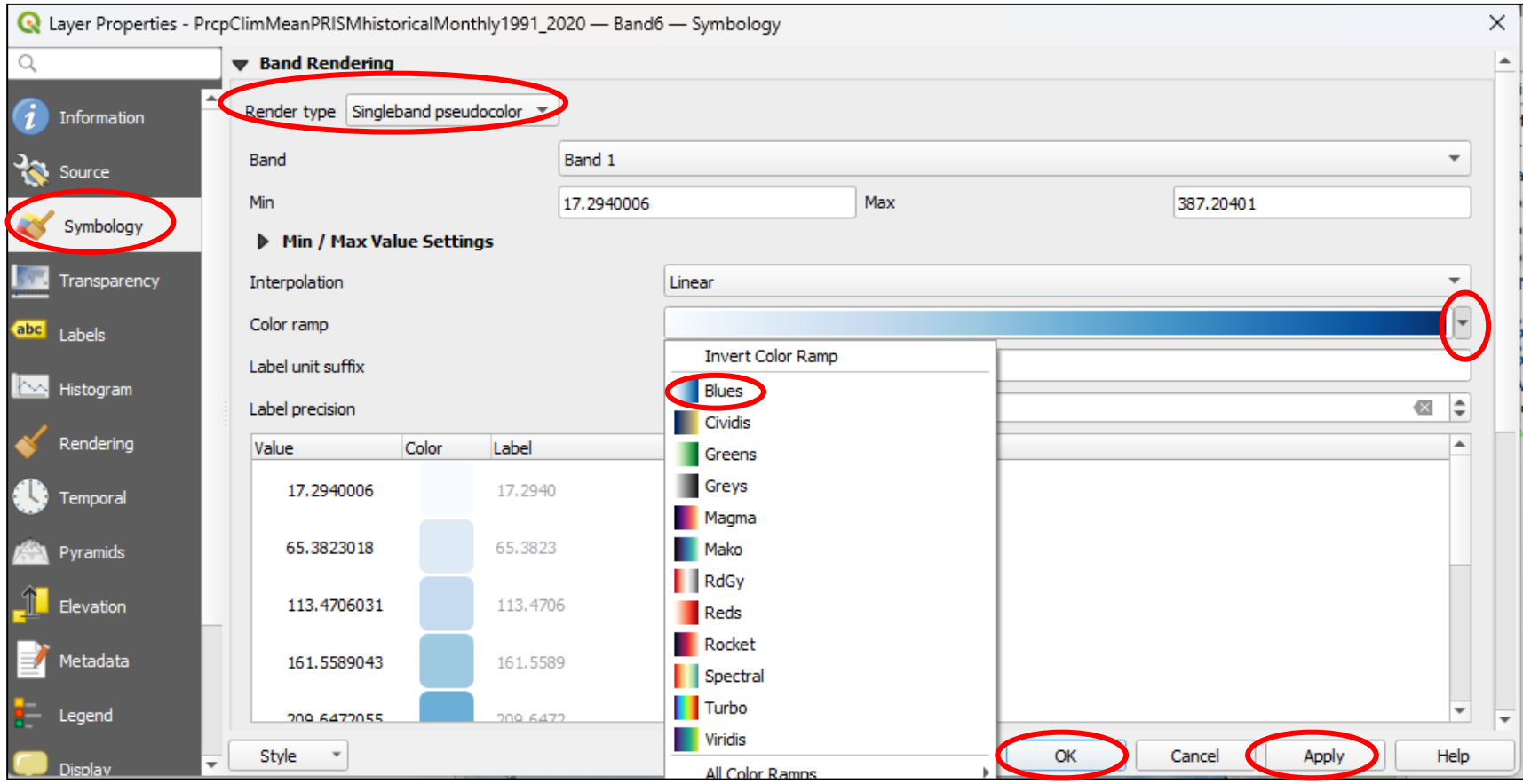




change NetCDF symbology (again)

- right-click on **PrcpClimMeanPRISMhistoricalMonthly1991_2020 — Band6,** select *Properties*
- under *Symbology,* change *Render type* to “Singleband pseudocolor” in drop-down
- use drop-down arrow beside *Color ramp* to choose “Blues”
- **Apply** and **OK**





navigate / examine raster layers

Navigate map with the *Pan* tool

Zoom in and out

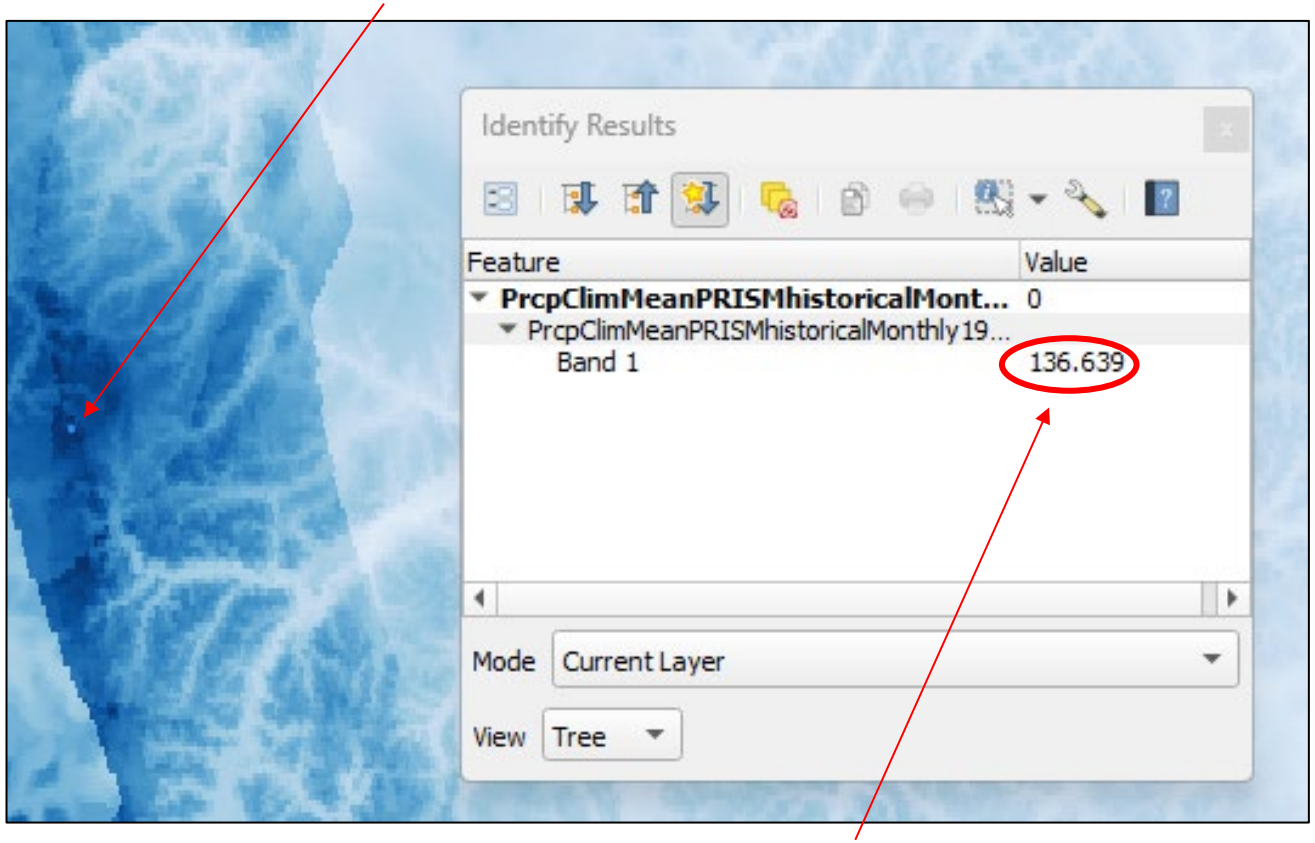
Use *zoom full* to see full data extent

select one of the NetCDFs in *Layers* panel

select the *Identify Features* tool
to examine the layer

Click on a pixel to display its precipitation amount (dark blue = more precipitation)

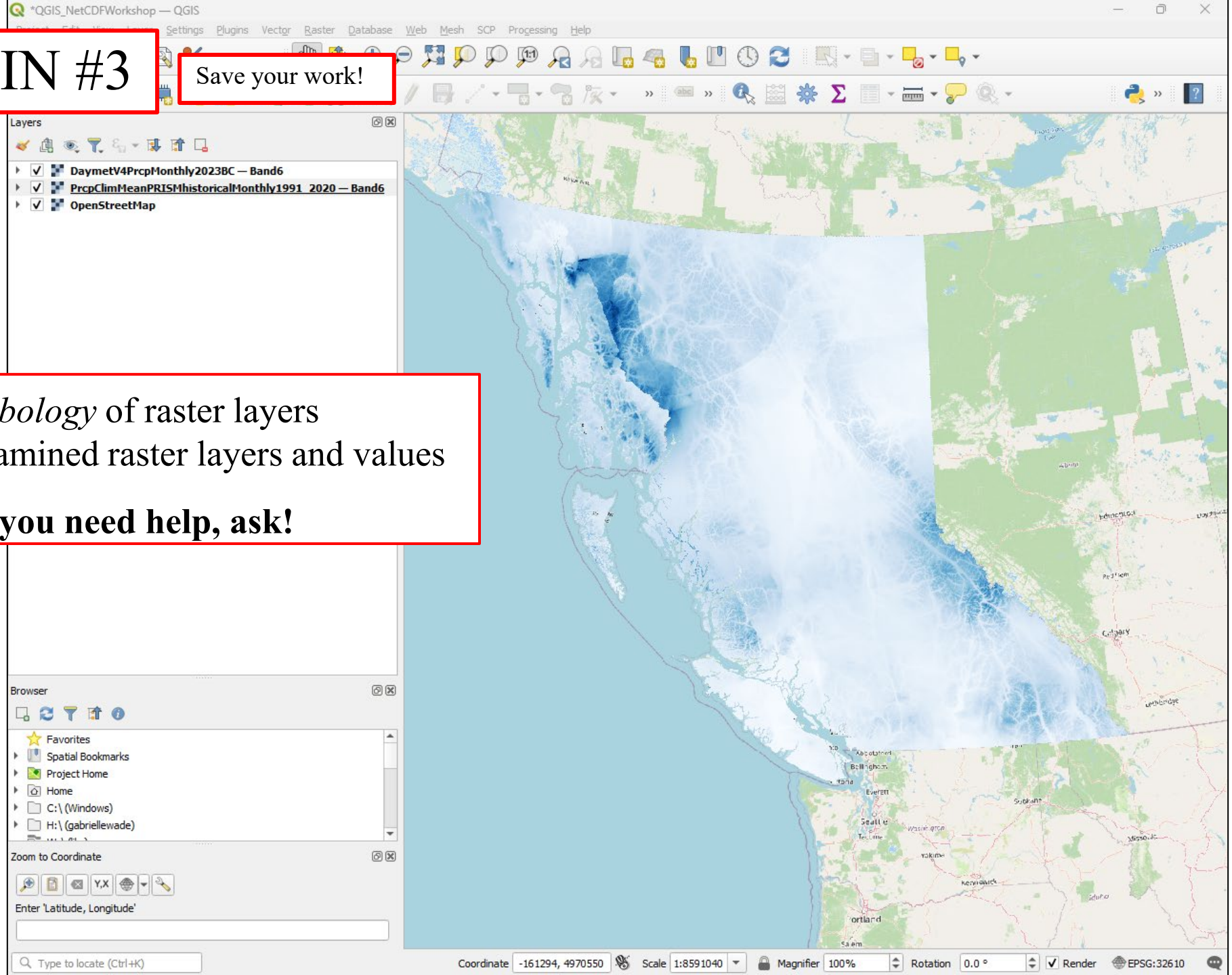
Selected pixel



1991-2020 normal June precipitation value (mm)

CHECK IN #3

Save your work!

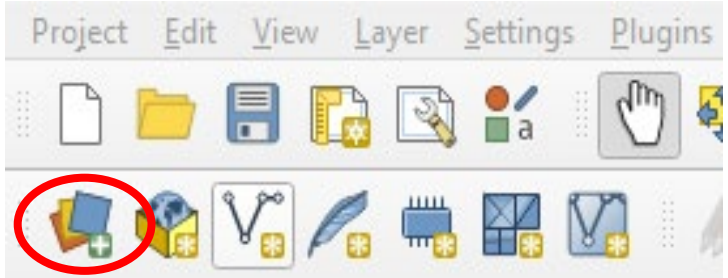


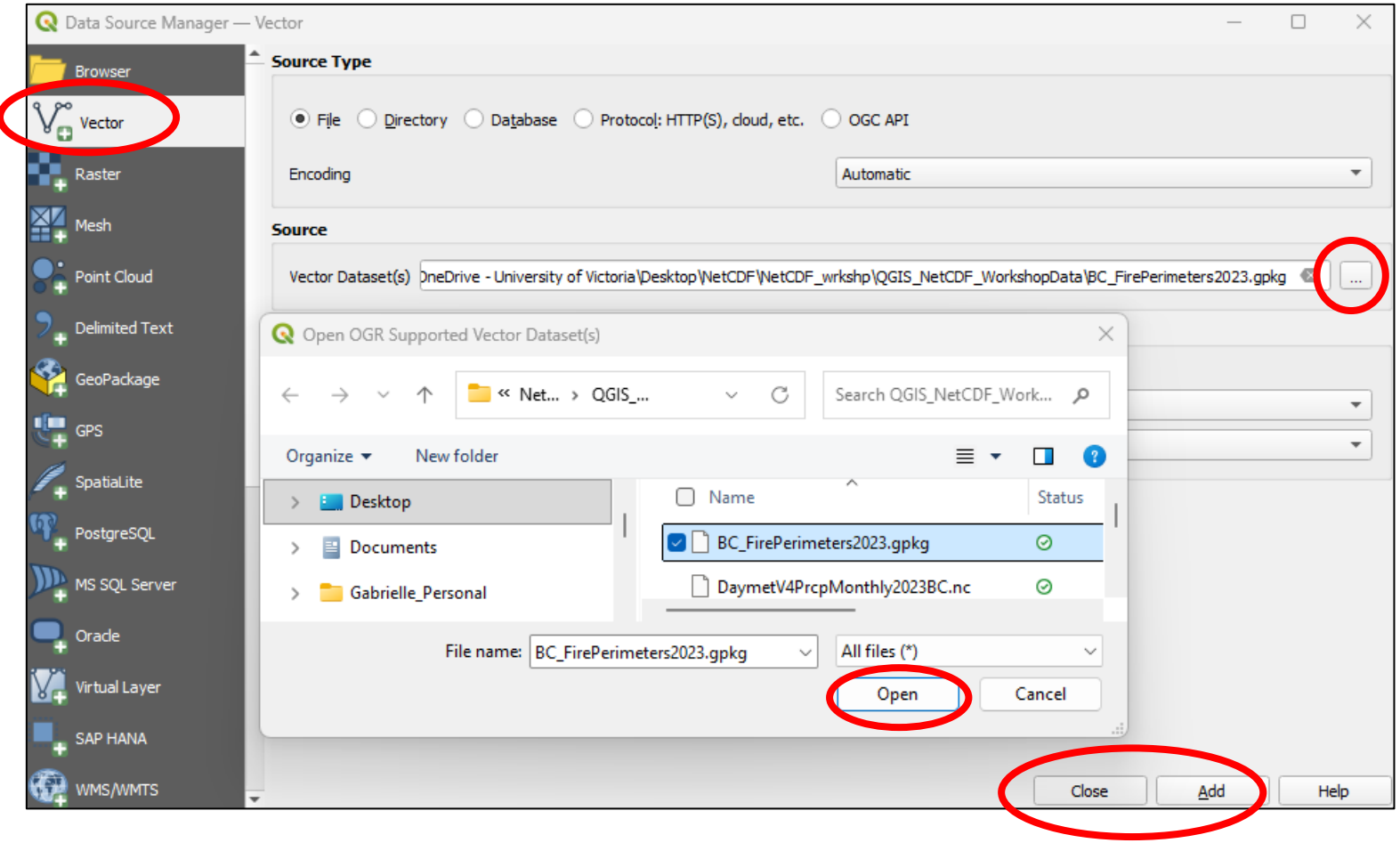
- changed *Symbology* of raster layers
- navigated/examined raster layers and values

If you need help, ask!

Activity #4

add BC_FirePerimeters2023





- Select *Open Data Source Manager*
- Select the *Vector* tab
- Under the "Source" heading click the
- Navigate to workshop data
- Select **BC_FirePerimeters2023.gpkg,** Open
- **Add** and **Close**

navigate / examine BC_FirePerimeters2023

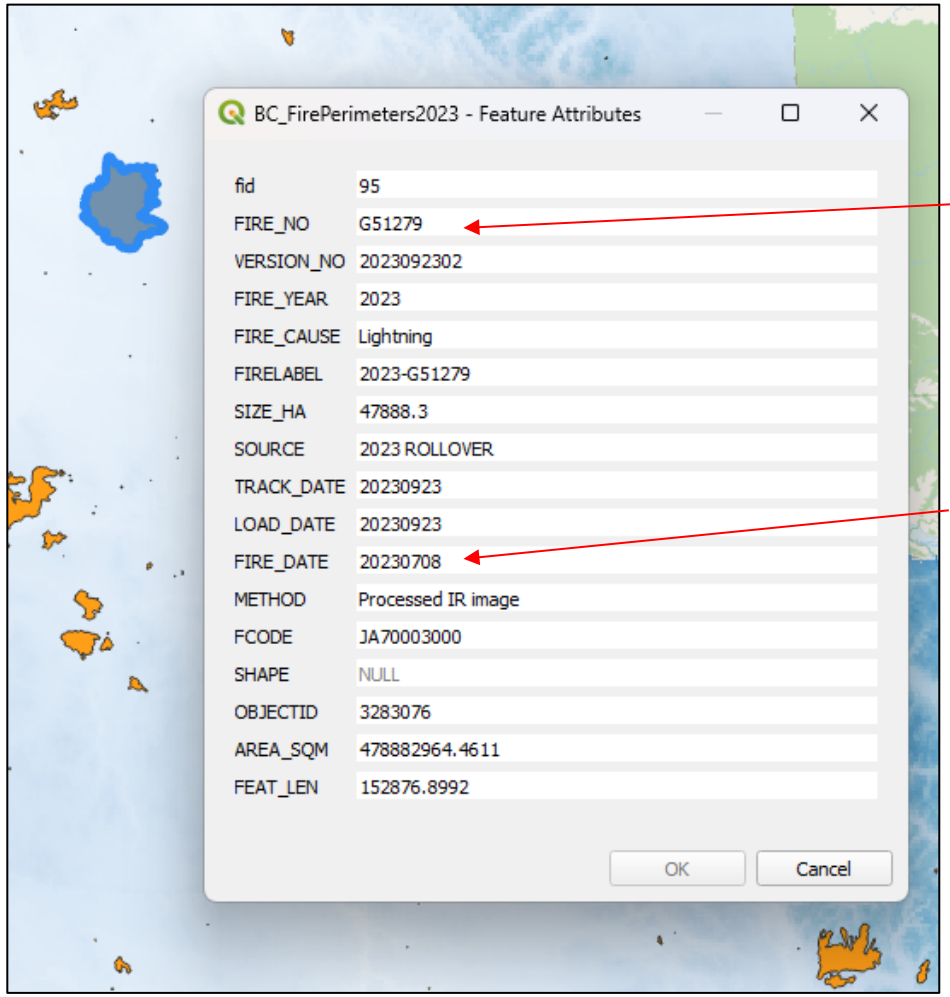
navigate map with the *Pan* tool

zoom in and out

select BC _FirePerimeters2023 in *Layers* panel

select the *Identify Features* tool
to examine the layer

click on a fire perimeter polygon to display its associated attributes

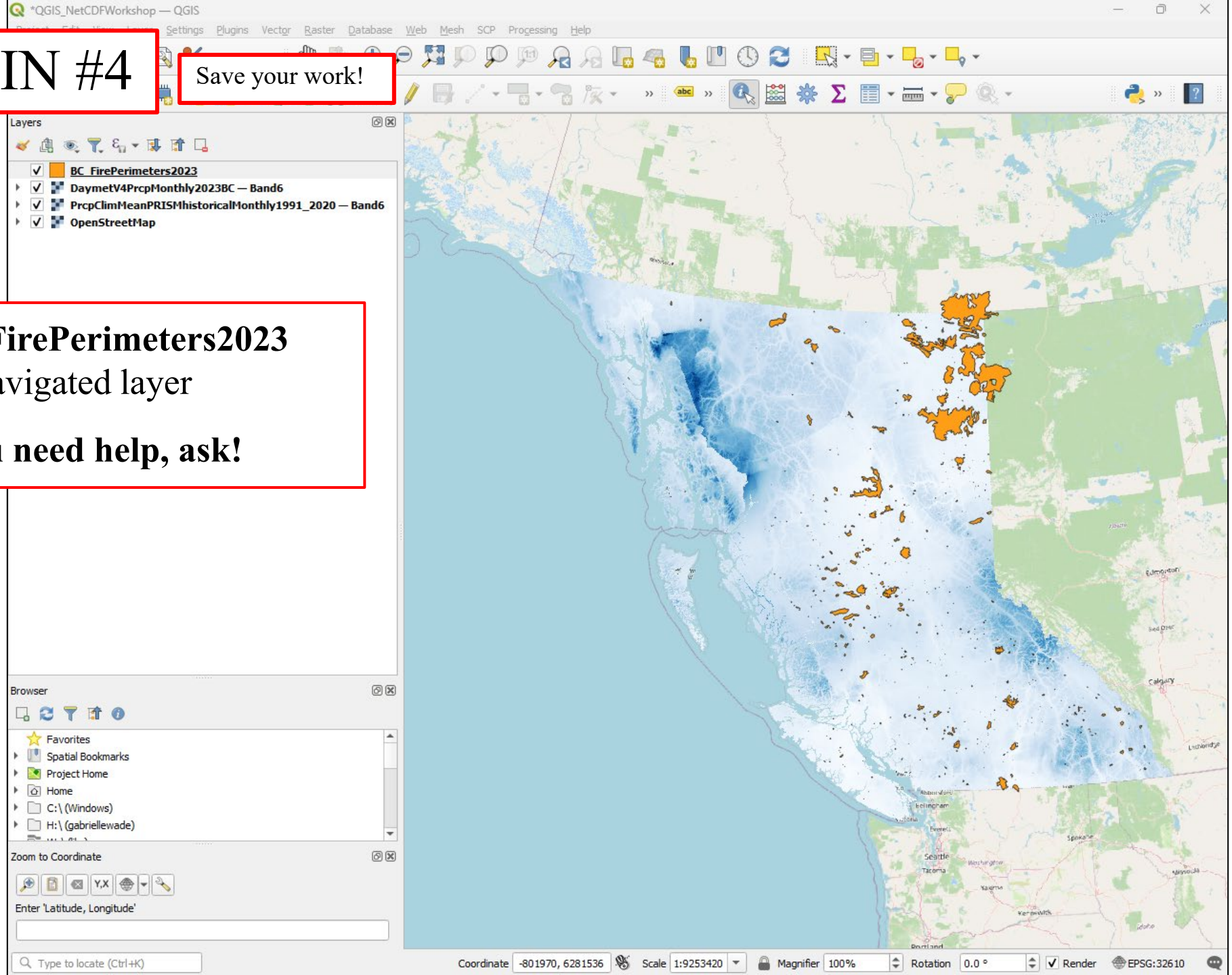


Fire_NO = G51279

Fire_date = July 8, 2023

CHECK IN #4

Save your work!



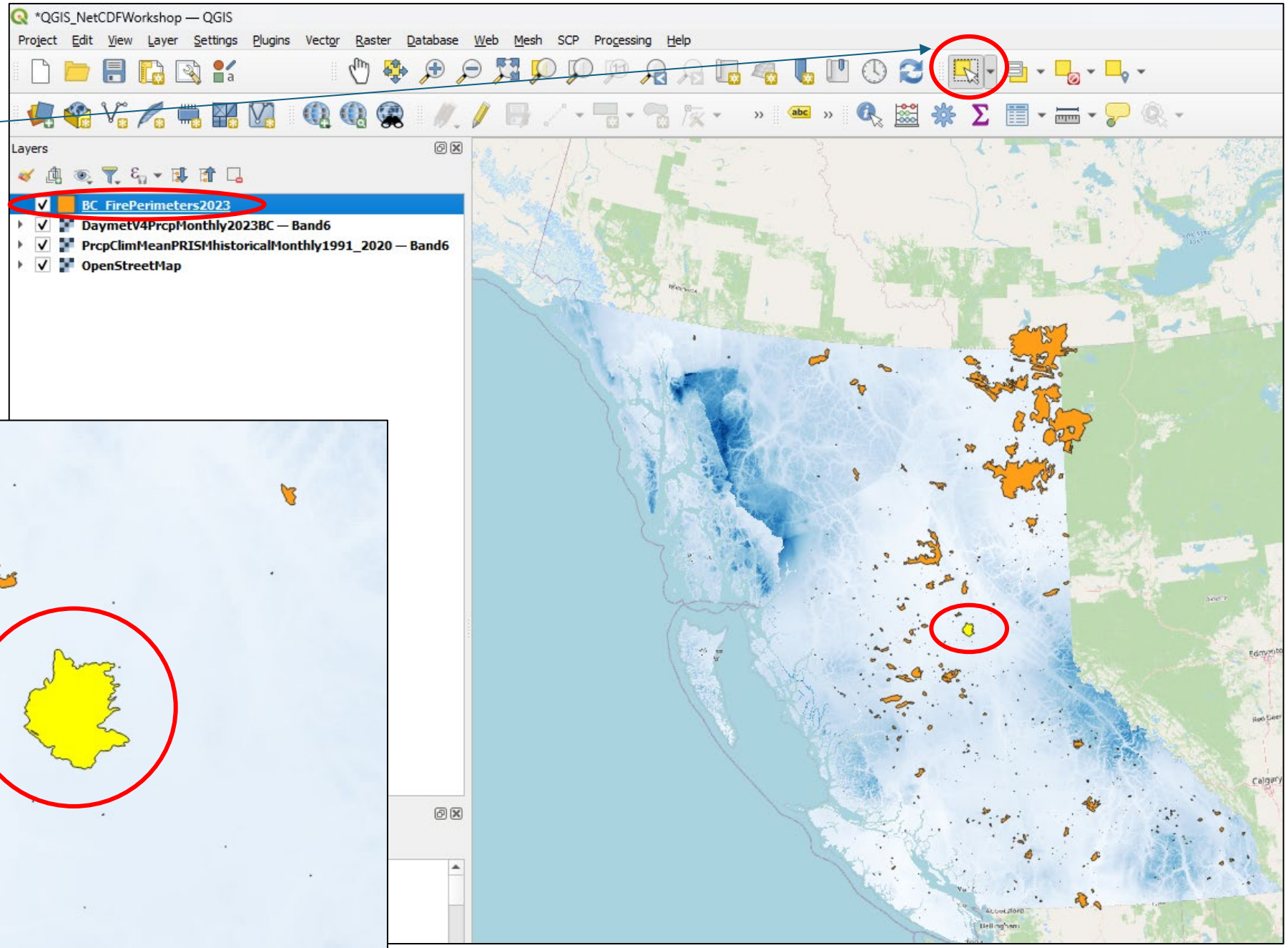
- added **BC_FirePerimeters2023**
- examined/navigated layer

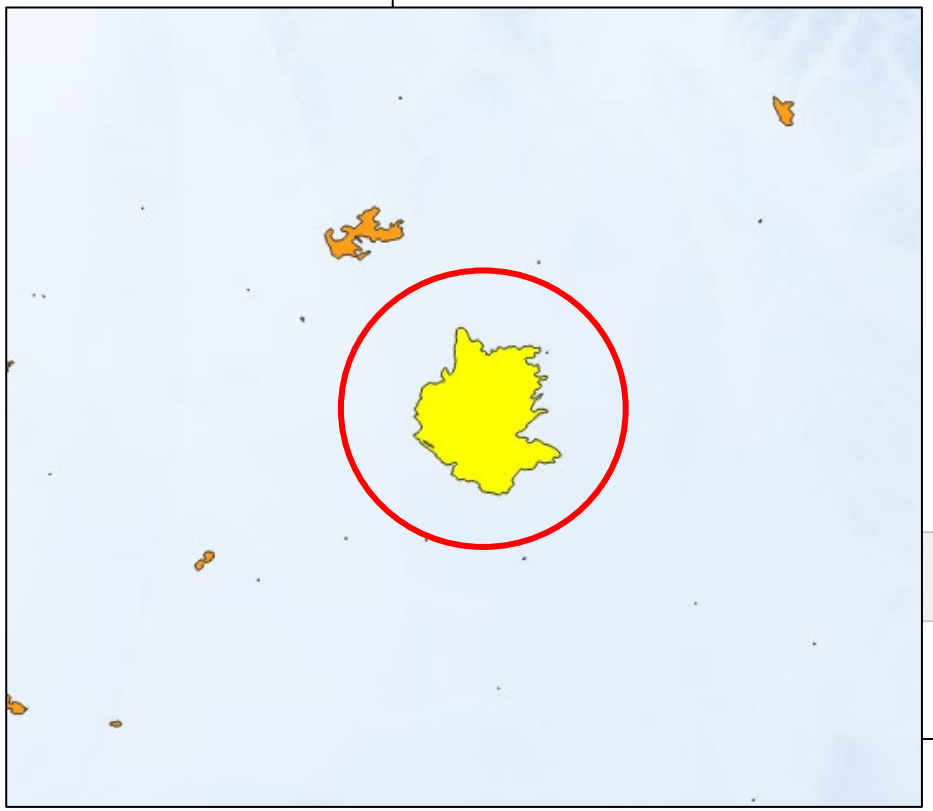
If you need help, ask!

Activity #5

select fire polygon

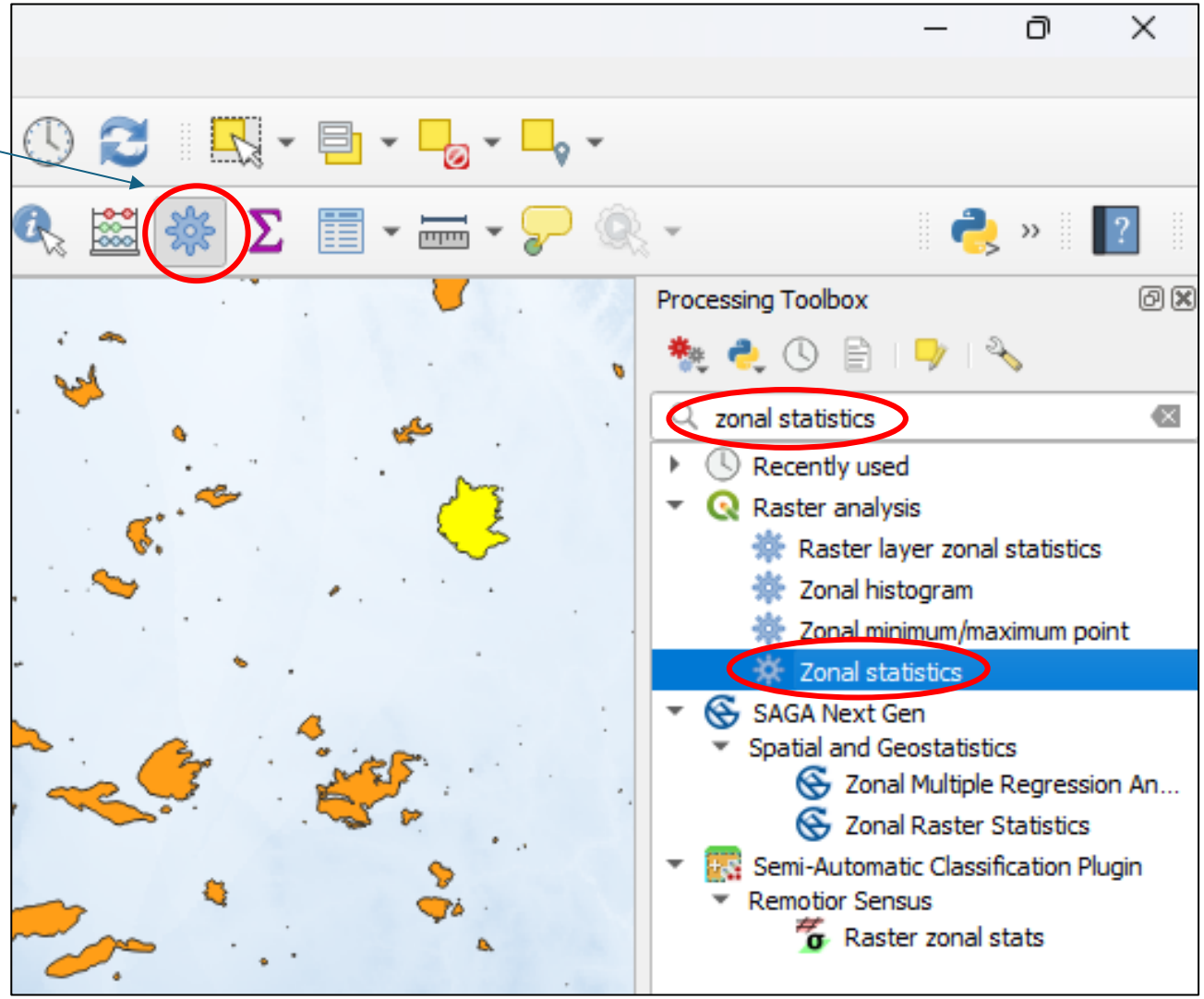
- choose *Select Features* tool from *Toolbar*
- Click **BC_FirePerimeters2023** in *Layers*
- Click on polygon shown in picture (central BC)
- Selected polygon will highlight in yellow





Zonal Statistics

- use *Zonal Statistics* tool to determine June 2023 precipitation for Fire G51279 polygon
- open *Processing Toolbox* from *Toolbar* and type ‘zonal statistics’
- double-click *Zonal Statistics* under *Raster Analysis*

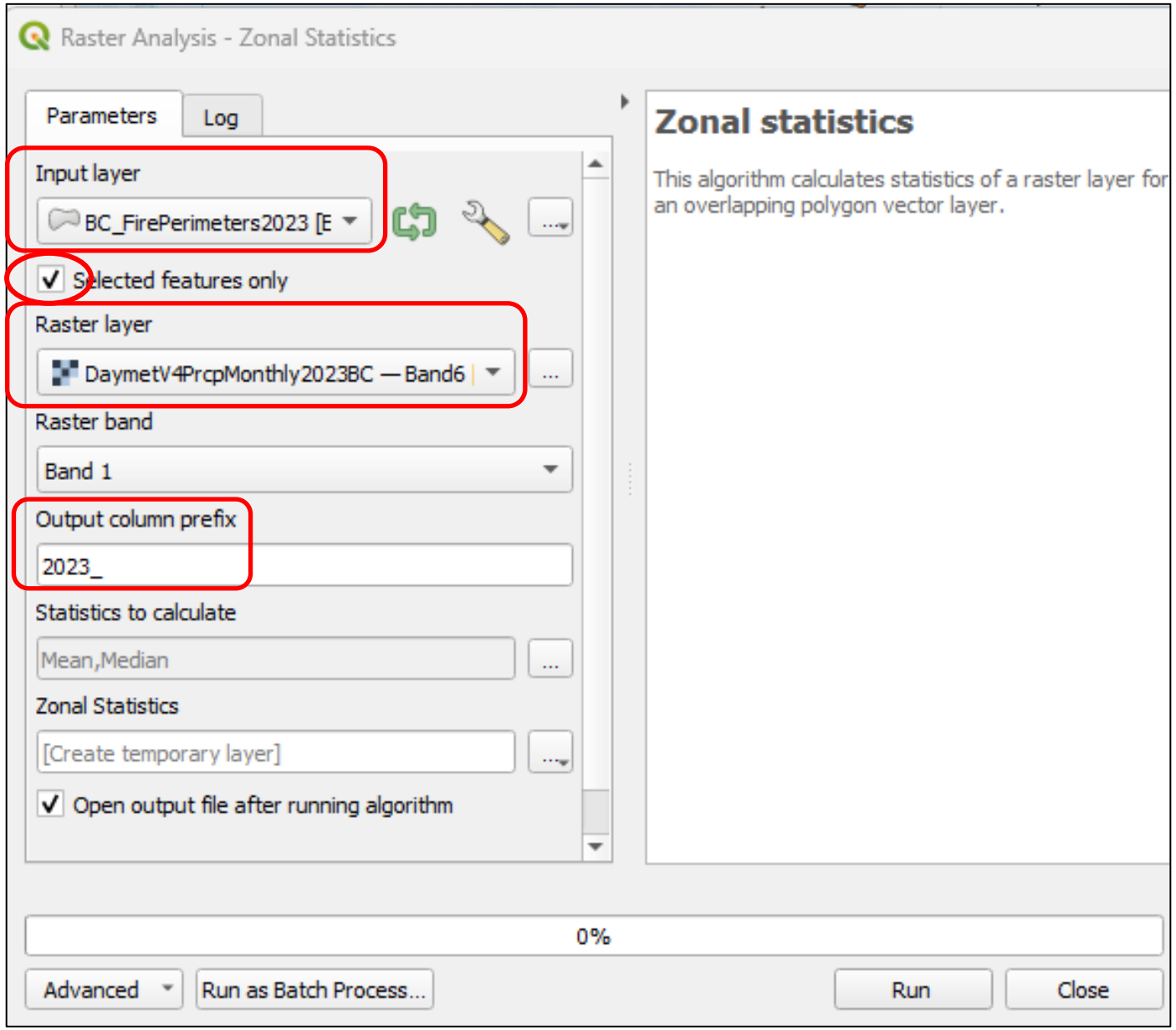


Zonal Statistics

- set *Input Layer* as **BC_FirePerimeters2023**
- check **Selected features only**
- set *Raster Layer* as **DaymetV4PrcpMonthly2023BC — Band6**

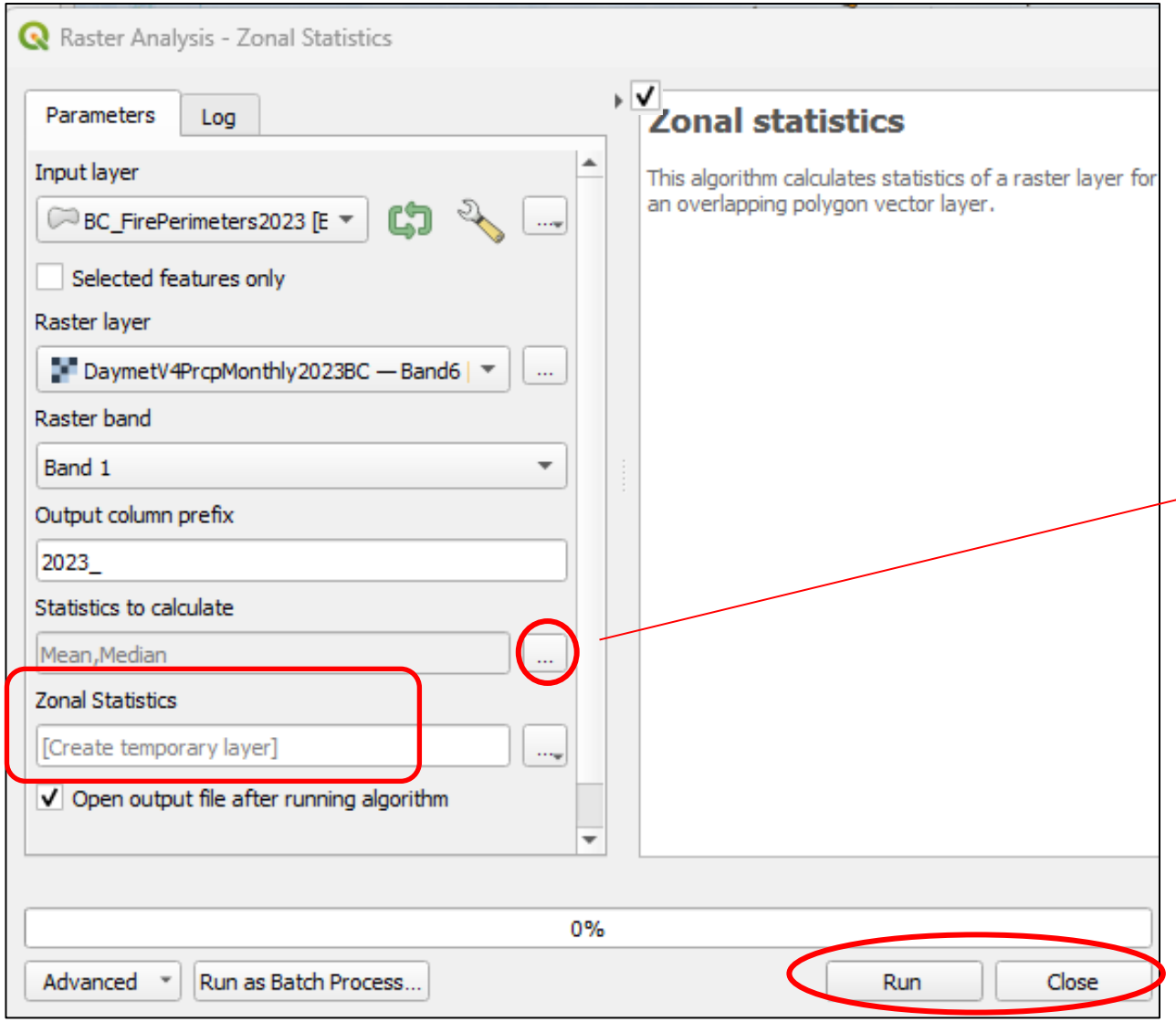
- change *Output column prefix* to 2023_

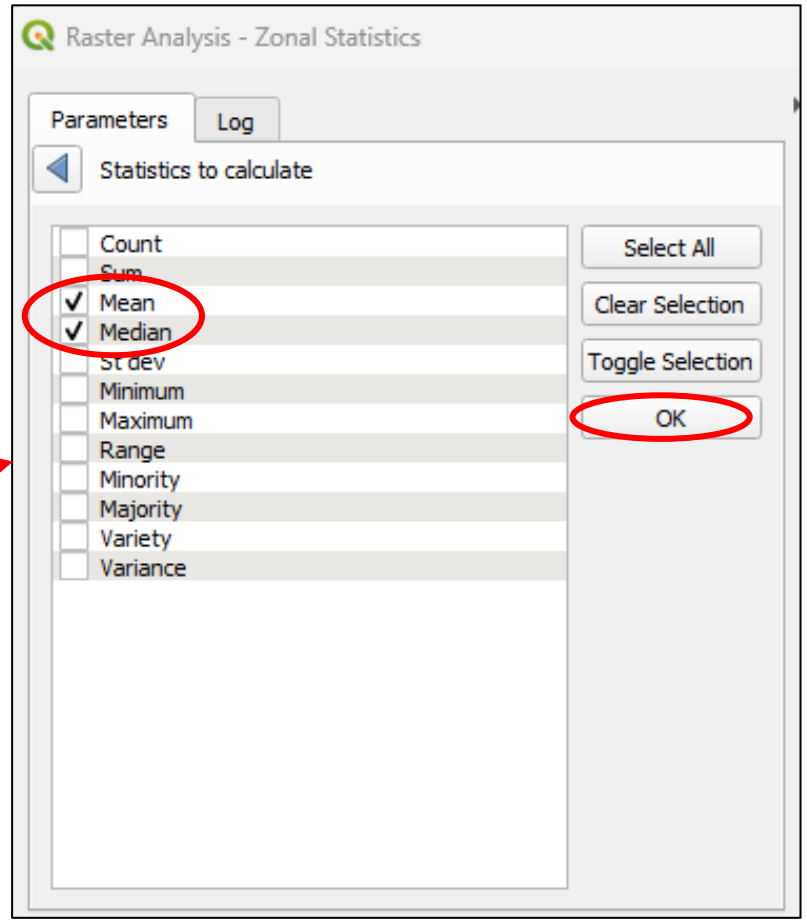
Continued on next slide..



Zonal Statistics

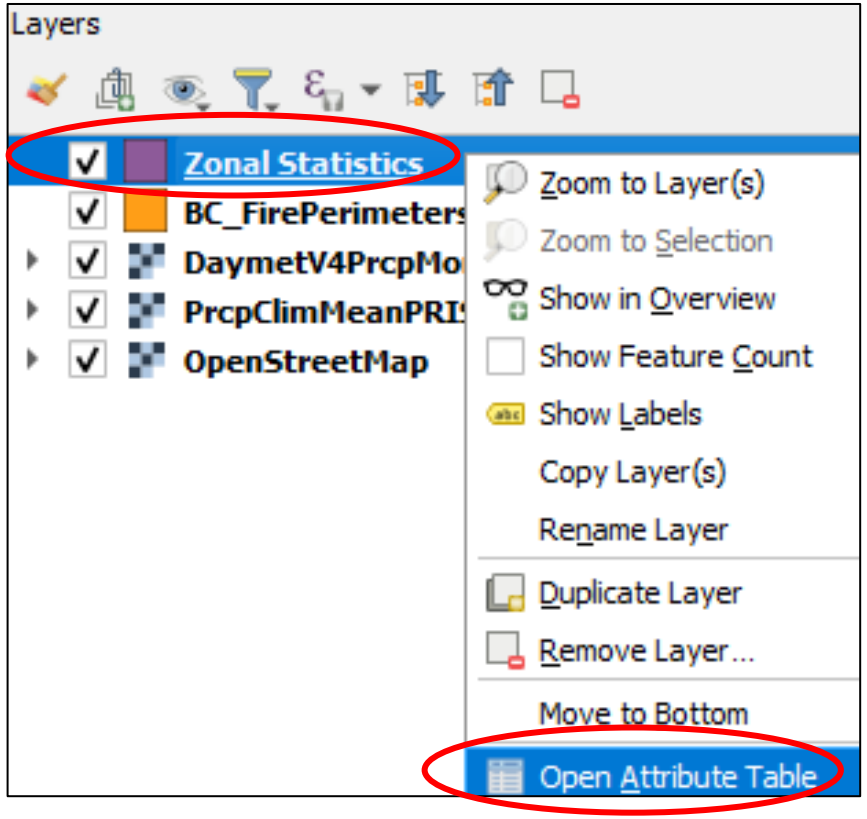
- click 3 dots ... beside *Statistics to calculate* and select only *Mean* and *Median;* ok
- Leave as *Temporary Layer* (could save to file…but won't today)
- **Run** and **Close**

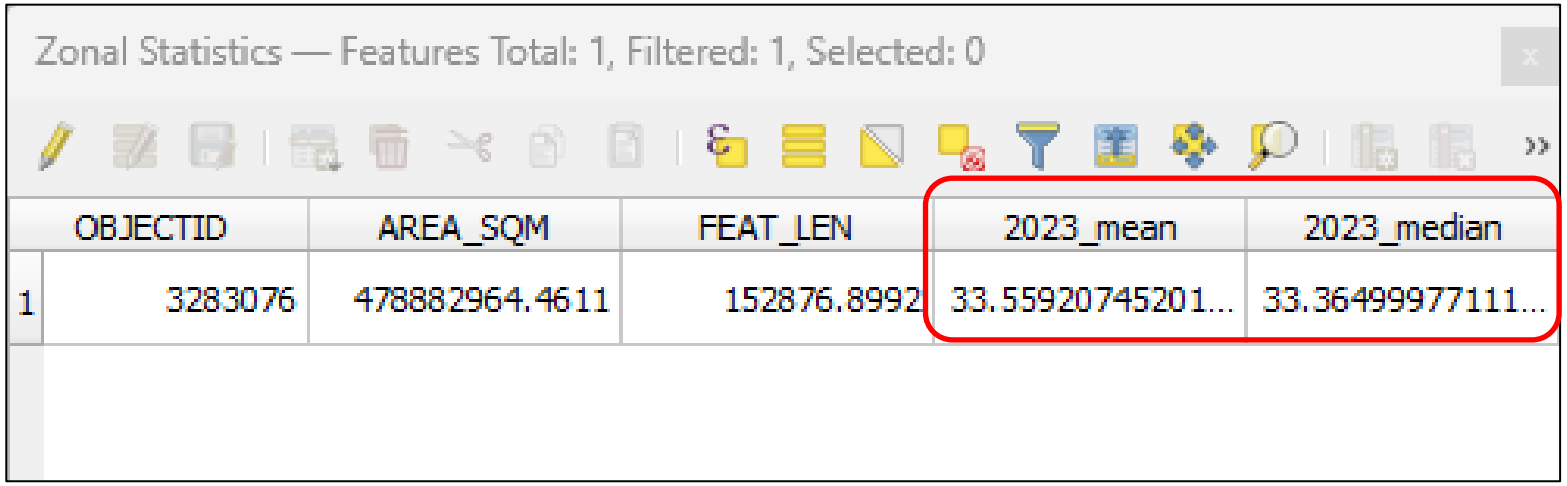




investigate *FireG80280 Perimeter_June2023*

- new **Zonal Statistics** layer will appear in *Layers* panel
- right click on **Zonal Statistics** and *Open Attribute Table*
- investigate attributes of fire…notice *Mean* and *Median* precipitation for this polygon in June 2023…
  (…will compare to June 1991-2020 precipitation…)





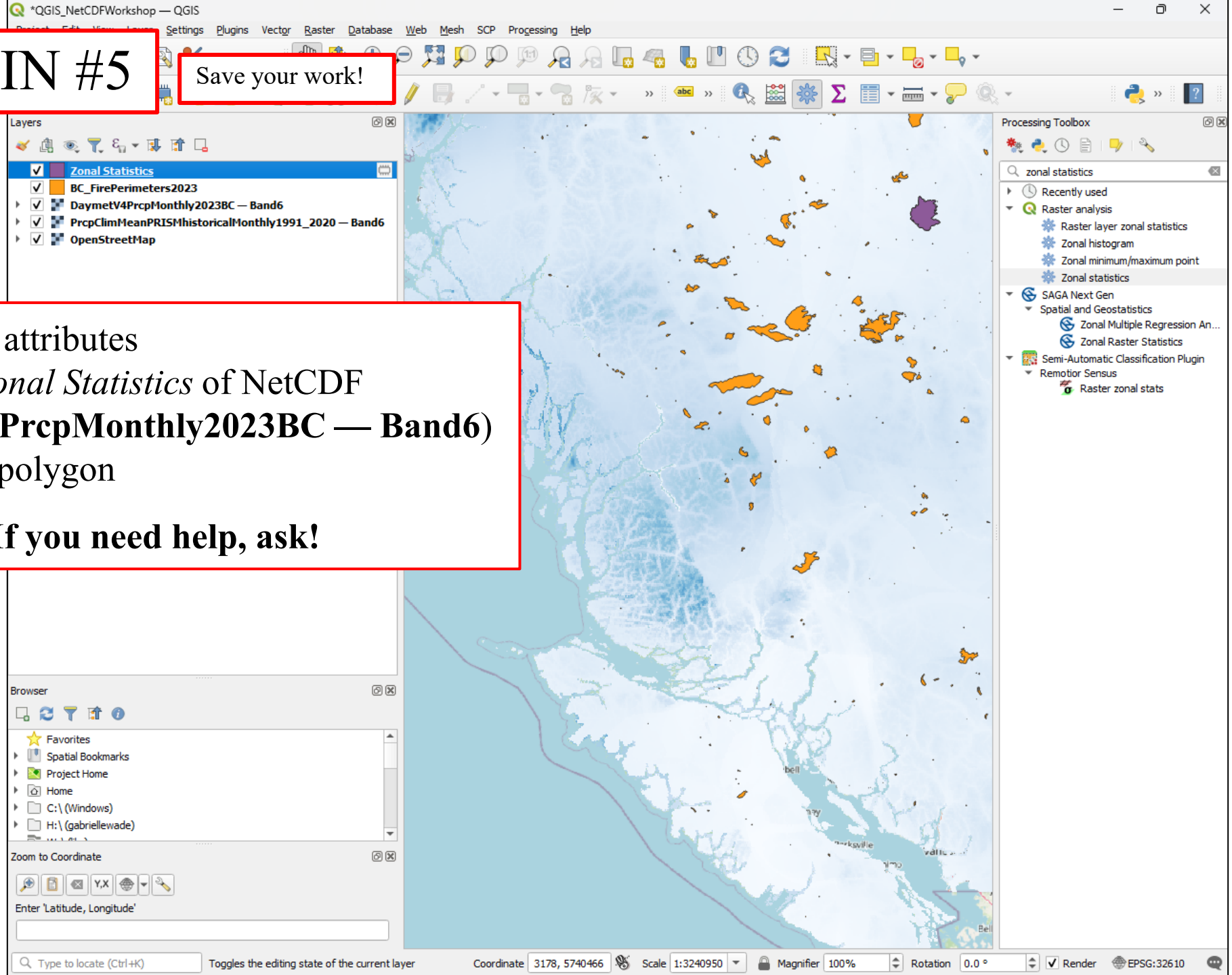
	OBJECTID	AREA_SQM	FEAT_LEN	2023_mean	2023_median
1	3283076	478882964.4611	152876.8992	33.55920745201...	33.36499977111...

CHECK IN #5

Save your work!

- investigated attributes
- generated *Zonal Statistics* of NetCDF (**DaymetV4PrcpMonthly2023BC — Band6**) for selected polygon

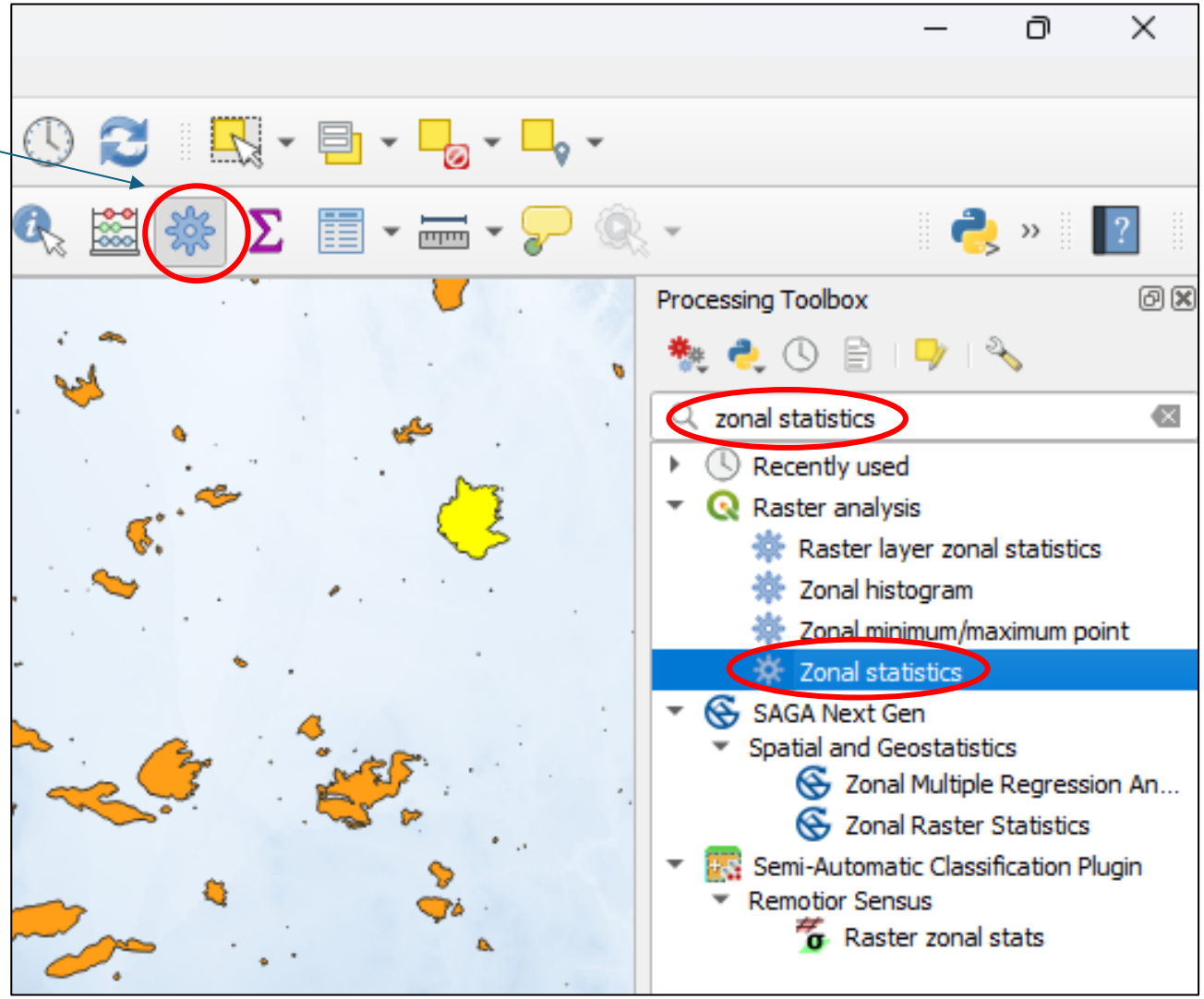
If you need help, ask!



Activity #6

Zonal Statistics (again)

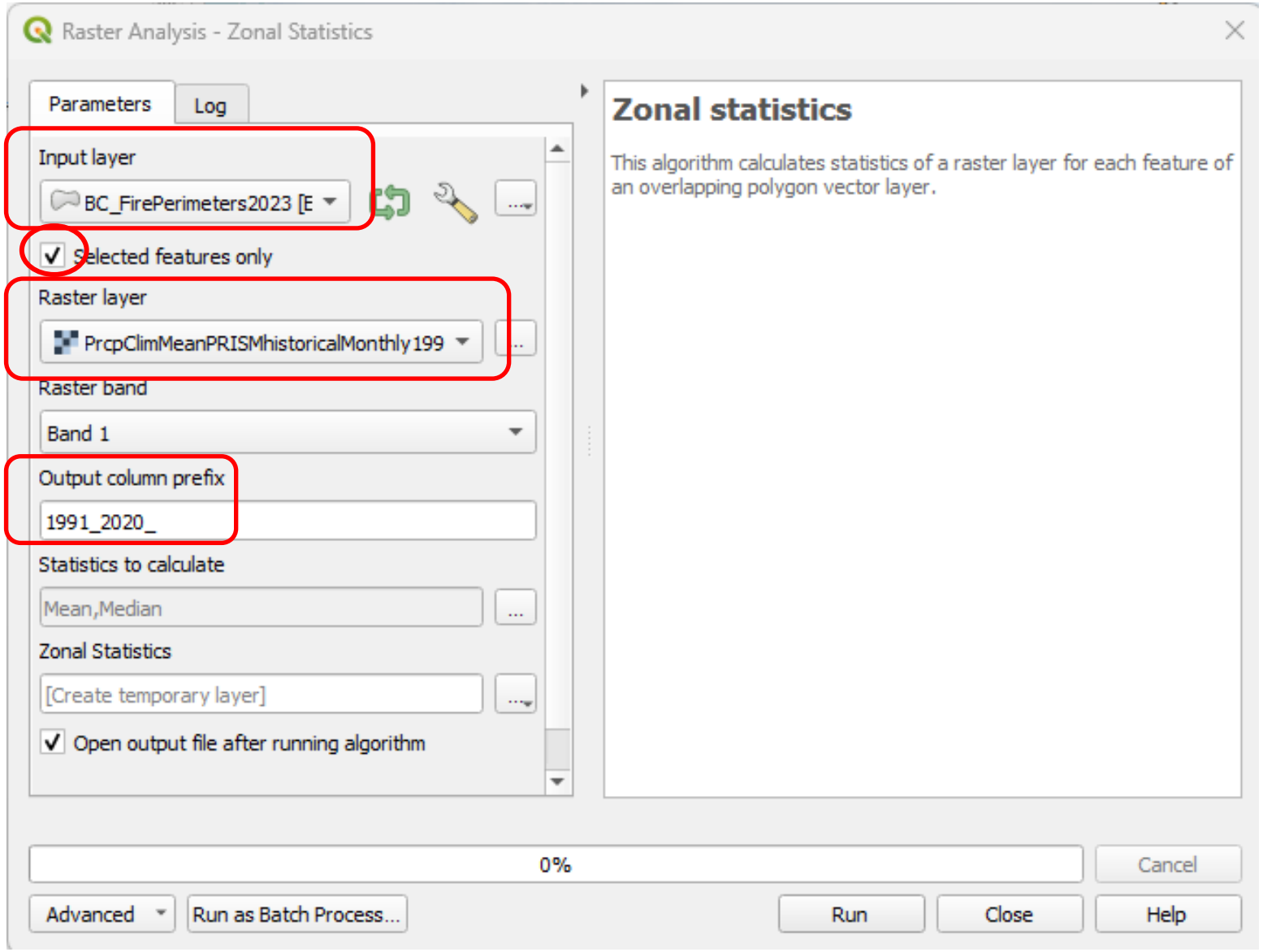
- use *Zonal Statistics* tool to determine June 1991-2020 precipitation for Fire G51279 polygon
- Open *Processing Toolbox* from *Toolbar* and type 'zonal statistics'
- Double-click *Zonal Statistics* under *Raster Analysis*



Zonal Statistics

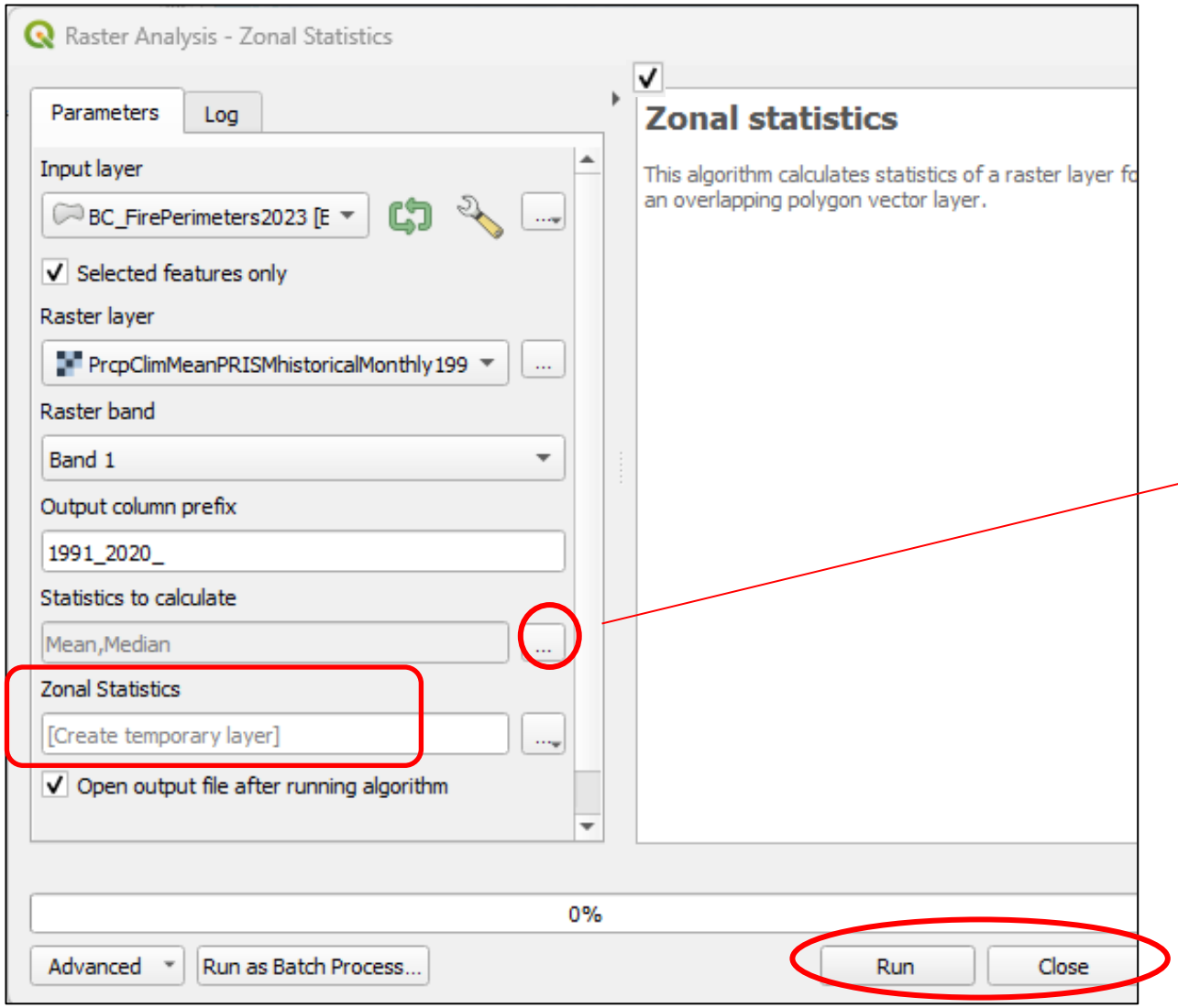
- set *Input Layer* as **BC_FirePerimeters2023**
- check **Selected features only**
- set *Raster Layer* as **PrcpClimMeanPRISMhistoricalMonthly 1991_2020 — Band6** (June)
- change *Output column prefix* to 1991_2020_

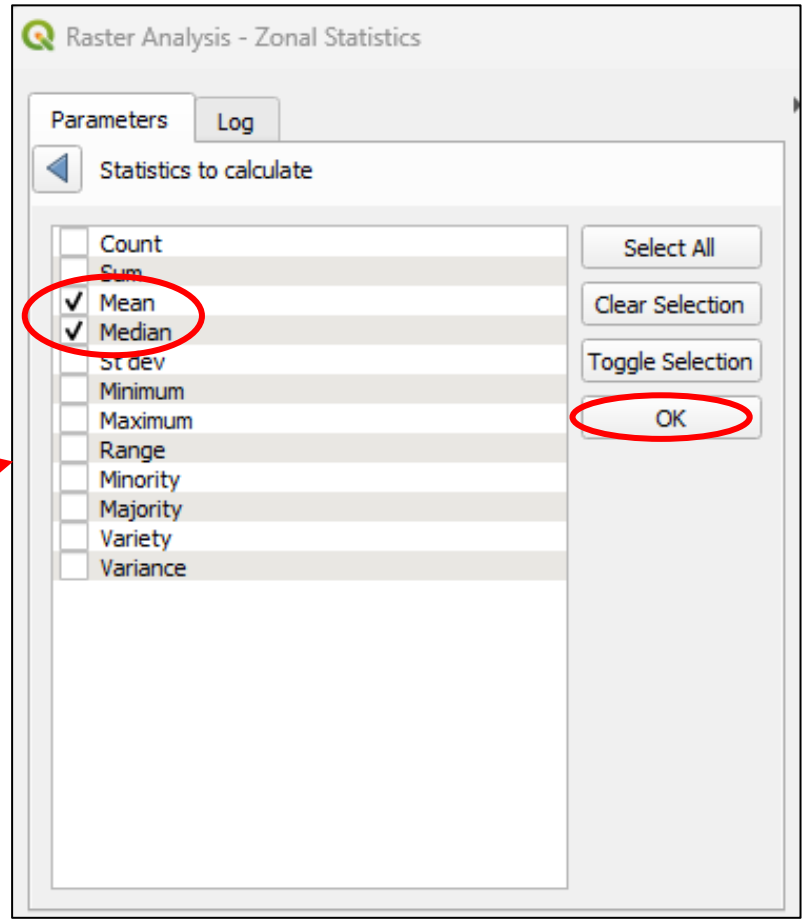
Continued on next slide..



Zonal Statistics

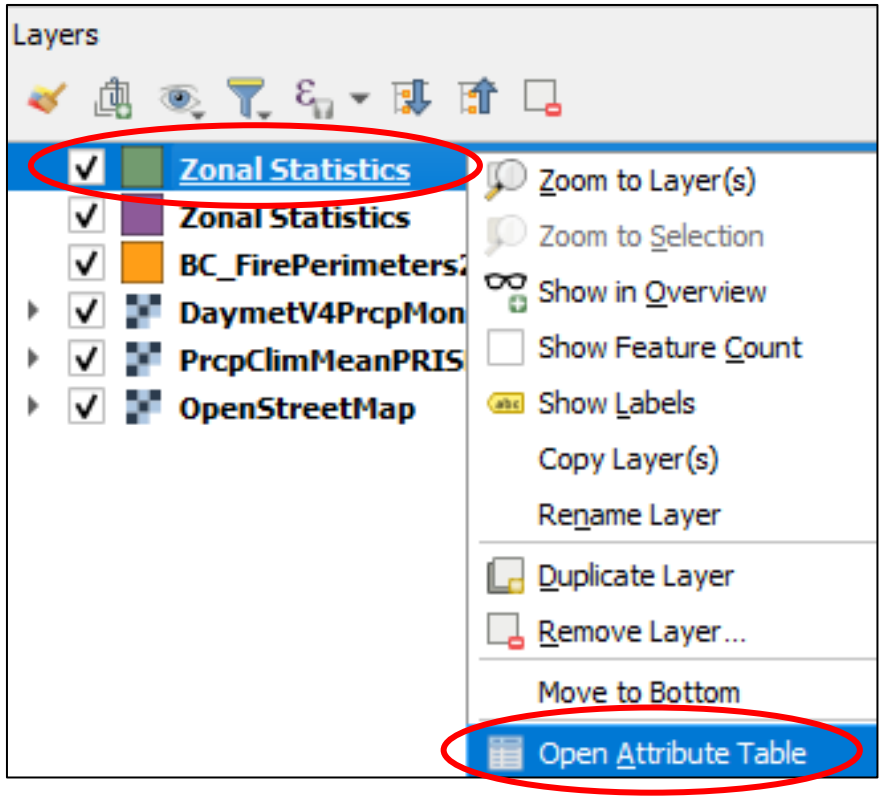
- click 3 dots ... beside *Statistics to calculate* and select only *Mean* and *Median;* ok
- Leave as *Temporary Layer* (could save to file…but won't today)
- **Run** and **Close**

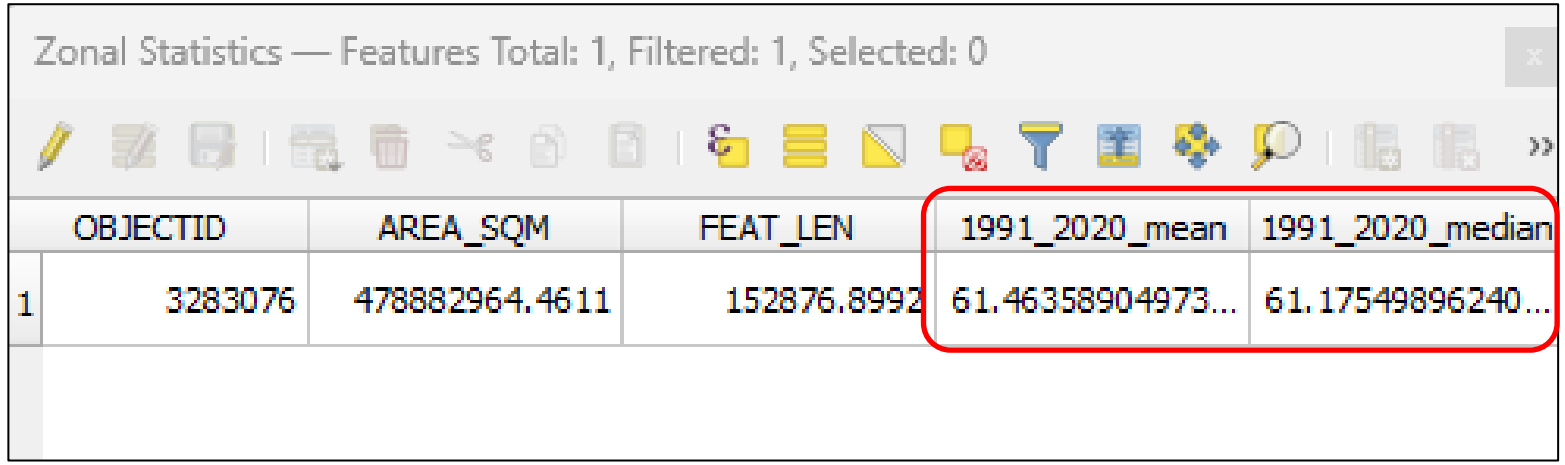




investigate *FireG80280 perimeter for June 1991-2020*

- new **Zonal Statistics** layer will appear in *Layers* panel
- right click on new **Zonal Statistics** and *Open Attribute Table*
- investigate attributes of fire…notice *Mean* and *Median* June precipitation (1991-2020) is almost double amount of precipitation in June 2023 (year of fire)…

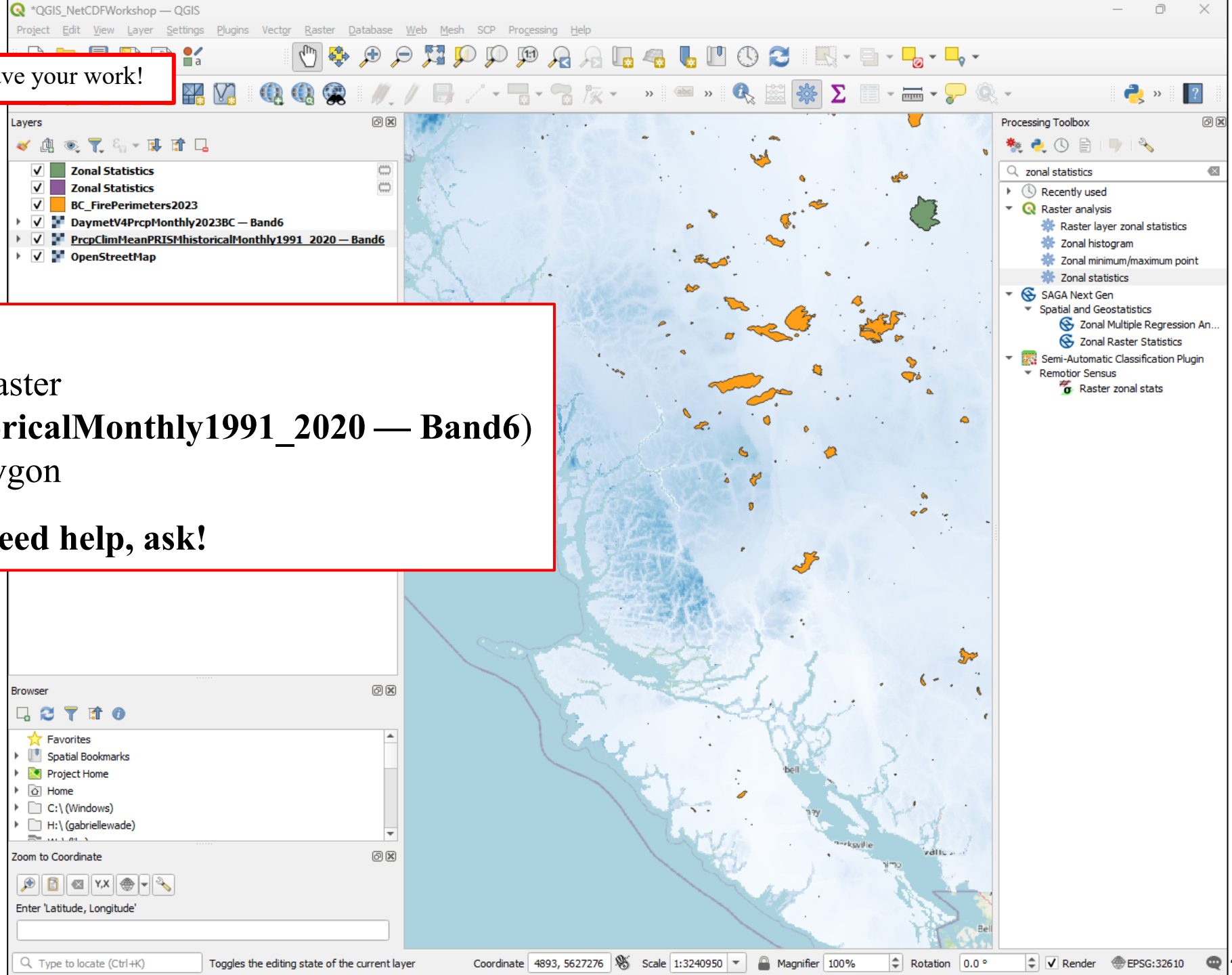




	OBJECTID	AREA_SQM	FEAT_LEN	1991_2020_mean	1991_2020_median
1	3283076	478882964.4611	152876.8992	61.46358904973...	61.17549896240...

CHECK IN #5

Save your work!



- investigated attributes
- generated *Zonal Statistics* of raster (**PrcpClimMeanPRISMhistoricalMonthly1991_2020 — Band6**) for selected fire perimeter polygon

If you need help, ask!

Congratulations!

You:

- loaded two different NetCDFs
- Symbolized NetCDF layers
- Examined NetCDF layers
- utilized vector polygons to determine precipitation rates in BC
  - (with 2023 *BC Fire Perimeters* and *QGIS* Zonal Statistics)

Resources going forward:

QGIS – additional resources:

- QGIS Training Manual: https://docs.qgis.org/3.44/en/docs/training_manual/index.html
- QGIS User Guide: https://docs.qgis.org/3.44/en/docs/user_manual/index.html
- QGIS Tutorials & Tips: https://www.qgistutorials.com/

Find data:

- GeoSpatial Data Guide: http://libguides.uvic.ca/geospatialdata

Questions or problems:

- UVic Geospatial Librarian (danielbm@uvic.ca) or YCW Geospatial Intern (gabriellewade@uvic.ca)

UVic full semester GIS courses in the Department of Geography:

- GEOG222 – Intro to Maps and GIS
- GEOG328 – GIS Analysis

GIS Skills and Mapping Micro-certificate (approx 30 hrs)

https://continuingstudies.uvic.ca/science-and-the-environment/programs/gis-skills-and-mapping